New Listings and Housing Market Congestion*

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Latest version: November 6, 2026

Abstract

This paper studies how newly listed homes affect properties already on the market through the search process. I develop a sequential-search model in which new listings enter buyers' search sequences and generate congestion that reduces incumbent visits and sale probabilities. Using detailed micro-level data, I find that a new listing reduces buyer visits to incumbent properties within the same building by 15–25% and their weekly sale probabilities by 11–27%, with no significant effects on listing or transaction prices. Congestion is stronger for more price-competitive entrants and closer substitutes, in smaller local markets, and among recently listed incumbents. Importantly, congestion persists despite joint buyer–seller entry, as new listings attract existing buyers. These findings provide micro-level evidence on the role of buyer search and matching in shaping housing-market congestion and liquidity.

Keywords: Housing Supply; Congestion Externality; Search and Matching; Stock–flow Matching

JEL Classification: R31, R21, D83

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1 Introduction

When a new property enters the housing market, how does it affect properties already on the market for sale? In a frictionless market, the arrival of an additional listing would be absorbed through adjustments in demand and prices. In a frictional housing market, however, search is costly: buyers inspect only a limited number of properties and therefore must rank available listings and decide which ones to visit. Moreover, housing search is highly segmented, so these choices are made within relatively narrow local markets and are directed by observable property characteristics (Albrecht et al., 2016; Han and Strange, 2016; Piazzesi et al., 2020). The arrival of a new listing can therefore reshape buyers' search processes. For example, a new listing can divert buyer attention away from incumbent properties, reducing their chances of being visited and sold. This search-based mechanism provides a micro-foundation for the seller-side congestion documented in housing markets (Anenberg and Ringo, 2026). The congestion issue is particularly important for home sellers. Housing is typically the largest asset on a household's balance sheet, while selling a home is an illiquid process with substantial uncertainty about both how long the property will remain on the market and the price at which it will eventually sell. Understanding how buyers and sellers search and match, and the micro-level mechanisms underlying congestion, is therefore essential to understanding housing-market liquidity.

To theoretically explain these congestion externalities, I develop a sequential-search framework in the spirit of Weitzman (1979). Unlike random matching, buyers rank properties in a search sequence and visit them sequentially according to this ranking. A newly listed property is inserted into buyers' search sequences and pushes incumbent properties to lower positions, reducing their probabilities of being visited and sold and thereby generating congestion. The model also predicts stronger congestion from more price-competitive entrants. A key implication is that congestion operates primarily through the reallocation of buyer attention across properties, with reduced visits serving as the central mechanism linking new-listing entry to lower incumbent sale probabilities.

This mechanism is difficult to observe with standard housing data, which typically record listings and transactions but not the underlying search process. As a result, there is limited direct evidence on whether and how new supply reshapes buyer search. I test these predictions using detailed micro-level data from a major Chinese housing platform that link property listings and transactions to buyer in-person visits. These data allow me to directly observe how buyer attention is reallocated following new-listing entry and distinguish between changes in buyer visits and changes in purchase probabilities, providing direct evidence on the model's

search mechanism.

I exploit variation in the timing of new-listing arrivals at the building and apartment complex levels, using difference-in-differences and event-study designs with rich sets of high-dimensional fixed effects to compare incumbent listings exposed to new entry with otherwise similar listings that are not. I estimate the effects of new-listing entry on incumbent buyer visits, sale probabilities, listing prices, and transaction prices. I also examine the buyer-side response to new-listing entry.

I find substantial negative congestion externalities from new-listing arrivals. Following the arrival of a new listing in the same building, existing properties receive approximately 15–25% fewer buyer visits and experience an 11–27% decline in their weekly probability of sale. Further analysis suggests that this decline is largely attributable to reduced buyer visits, with limited evidence of a change in purchase decisions conditional on visits. This indicates that congestion operates mainly through the reallocation of buyer attention. The congestion externality also extends beyond the building itself: new listings reduce the sale probability of existing properties in other buildings within the same apartment complex. Despite the substantial decline in liquidity, I find no significant price response. Existing sellers neither adjust their listing prices nor sell at lower transaction prices following the arrival of a new listing. Thus, the congestion effect operates primarily through the liquidity margin rather than the price margin.

I find substantial heterogeneity in congestion effects. First, more price-competitive entrants generate stronger congestion, consistent with the sequential-search model's prediction that they are more likely to move ahead of incumbents in buyers' search sequences. Second, congestion is concentrated among recently listed incumbents, consistent with stock–flow matching: recent listings compete for the existing stock of buyers, whereas older listings increasingly rely on newly arriving buyers. Third, congestion is stronger when properties are closer substitutes, particularly when they share the same floor or floor area. Finally, congestion is weaker in larger markets, where an additional listing represents a smaller expansion of the available choice set.

New-listing entry may also induce a demand-side response, leading to joint buyer–seller entry. I therefore examine how new listings affect local buyer activity and the role of stock–flow matching in shaping congestion. I find that new listings attract additional buyers into the building, increasing both total visits and market tightness at the building level. Yet incumbent properties receive fewer visits and become less likely to sell. Following new-listing entry, buyers visiting the building exhibit a temporary increase in their average time on market (TOM) at the time of their visits, suggesting that some of the additional activity comes from buyers who have already been searching for some time. This pattern is consistent with

stock–flow matching: new listings attract the existing stock of searching buyers, while older incumbents increasingly rely on newly arriving buyers. These findings help explain why local buyer activity can increase even as congestion reduces incumbent liquidity.

Related Literature. This paper is most closely related to Anenberg and Ringo (2026), who document congestion in the U.S. housing market. My findings are consistent with theirs in showing that congestion operates primarily through lower sale probabilities rather than lower prices. I complement their analysis in three ways. First, I develop a sequential-search model that provides a micro-foundation for congestion by showing how new listings can push incumbent properties to lower positions in buyers' search sequences. Second, I directly examine the buyer-visit margin and show that new-listing entry reduces incumbent sale probabilities primarily by diverting buyer attention away from existing properties, consistent with the model's search-ranking mechanism. Third, I examine joint buyer–seller entry and find that congestion persists even as new listings attract additional buyers into the local market. Consistent with stock–flow matching, these additional buyers tend to have longer time on market and direct their attention toward newly listed properties rather than incumbents. Together, these findings provide a micro-level foundation for housing-market congestion and show how the allocation of buyer search shapes its incidence across incumbent sellers.

This paper contributes to the literature on the local effects of changes in housing supply. Existing studies show that new housing supply can affect nearby rents and house prices through changes in supply, competition, and neighborhood composition, with mixed findings. Li (2021), Asquith et al. (2023), and Mense (2025) find that new housing supply lowers nearby rents, whereas González-Pampillón (2022) documents increases in nearby house prices associated with reductions in crime and increases in neighborhood income. Other work documents highly localized spillovers generated by changes in the quality or availability of nearby housing, including renovation, rehabilitation, and foreclosure (Anenberg and Kung, 2014; Campbell et al., 2011; Ganduri and Maturana, 2024; Hartley, 2014; Rossi-Hansberg et al., 2010; Szumilo, 2020). I contribute to this literature by studying local supply changes through the lens of search frictions. Rather than operating primarily through prices, I find that the arrival of new listings reallocates buyer attention and reduces the liquidity of incumbent properties: existing listings receive fewer visits and become less likely to sell, while listing and transaction prices respond little. These findings highlight a distinct channel through which local housing supply affects incumbent sellers, operating through buyer search and matching rather than the price margin.

This paper also contributes to the literature on how the joint entry and exit of buyers and sellers shape housing-market dynamics. Search-and-matching models emphasize market tightness—the ratio of buyers to sellers—as a central determinant of matching rates and market liquidity (Genesove and Han, 2012; Han and Strange, 2015). Empirically, Genesove and Han (2012) show that stronger demand reduces seller time on market, buyer time on market, and the number of properties visits, and use these responses to estimate a housing matching function. A related strand of work examines how changes in buyer and seller entry jointly shape housing-market activity and decomposes the distinct contributions of each side of the market (Anenberg and Ringo, 2024; Ngai and Sheedy, 2024). In addition, the sequencing of buying and selling decisions can affect housing-market dynamics by endogenously altering market tightness (Grindaker et al., 2026; Moen et al., 2019; Ngai and Sheedy, 2020). I contribute to this literature by moving beyond the random-matching framework to examine how new-listing entry affects both buyer activity and incumbent sellers. I show that congestion persists even with joint buyer–seller entry: new listings attract additional buyer attention and increase local market tightness, yet incumbent properties receive fewer visits and become less likely to sell. These findings highlight how the reallocation of buyer attention can generate congestion externalities even as local buyer activity expands.

This paper also relates to the broader literature on search and matching in housing markets, particularly recent work using granular platform search data to directly observe buyer search behavior and the matching process. Related work on directed search shows that buyers allocate search toward properties based on observable listing characteristics, particularly asking prices (Albrecht et al., 2016; Han and Strange, 2016). Another strand emphasizes stock–flow matching, whereby newly entering buyers and sellers interact differently with market participants that have remained unmatched (Coles and Smith, 2026; Gilbukh, 2023; Smith et al., 2026). I provide descriptive evidence of stock–flow matching and show how this mechanism helps explain congestion effects. An alternative framework for understanding search and matching in markets is sequential search (Armstrong, 2017; Weitzman, 1979). Building on Weitzman (1979), I develop a sequential-search framework that explains how new listings reduce incumbent visits and sale probabilities and predicts which properties are most exposed to these congestion effects. More broadly, a growing literature uses detailed search data to study how buyers allocate attention across housing options. Gargano et al. (2023) document how online housing search responds to local house-price experiences, while Piazzesi et al. (2020) emphasize the segmentation of housing search across locations and property characteristics. van Dijk and Francke (2018) show that online interest in listed properties contains information about housing demand and predicts market

liquidity and prices. My analysis contributes to this literature by using detailed search data from a housing brokerage platform to directly observe how the arrival of new supply reallocates buyer attention across competing properties and affects seller liquidity.

The rest of the paper is organized as follows. Section 2 presents a model that motivates the empirical findings. Section 3 introduces the data and empirical strategy. Section 4 presents and discusses the empirical results. Section 5 concludes.

2 A Sequential Search Model

I use a simple static sequential-search model to illustrate how the arrival of a new listing generates congestion for properties already on the market. Buyers face a fixed set of properties and inspect them sequentially according to an optimal search rule. The arrival of a new listing perturbs this choice set and changes the ranking of incumbent properties in buyers' search sequences. By shifting some incumbent properties to lower positions, the shock reduces their probabilities of being visited and of being sold. The model therefore provides a comparative-static framework for analyzing how new-listing entry affects incumbent sellers through the search process.

2.1 Market Structure

Rather than modeling the full market dynamics, I focus on a local cross-section of the market around the arrival of a new listing. Just before that arrival, there is a finite set of incumbent buyers $\mathcal{B}$ and incumbent sellers $\mathcal{S}$, where there are fixed number of buyers $|\mathcal{B}| = B$ and fixed number of sellers $|\mathcal{S}| = S$. Each seller $k \in \mathcal{S}$ lists one property for sale, and each buyer $i \in \mathcal{B}$ searches over the set of currently available properties.

The new listing enters the market as an unanticipated shock.[1] Before entry, the new listing's characteristics and potential value to buyers are unknown and are modeled as random draws from the same distributions as those of incumbent properties. Once the new listing enters the market, its characteristics and potential value to buyers become known, and it is added to the set of available properties. At that point, incumbent buyers may be at different stages of their search: some may not yet have visited any property, while others may already have inspected

[1] In the model, new-listing entry is treated as an exogenous shock that buyers cannot anticipate or incorporate into their search strategies. Otherwise, the original search strategies in Weitzman (1979) would no longer be optimal.

an arbitrary number of listings. Likewise, incumbent properties may have accumulated different numbers of buyer visits. I take this cross-sectional distribution of ongoing search histories as given at the moment of entry and study how the entrant changes subsequent search and matching outcomes.[2]

2.2 Weitzman Search

I build on the sequential search framework of Weitzman (1979), who studies the optimal search problem of a single decision maker choosing among multiple alternatives with costly information acquisition. I extend this environment to a market with multiple buyers and sellers by allowing each buyer to solve an individual Weitzman search problem over the common set of available properties following Ursu et al. (2025).

For each property $k \in \mathcal{S}$, buyer i observes its list price p_k and a set of pre-search observable characteristics summarized by a prior value v_{ik}. The prior value is known to the buyer when she forms her search strategy, varies across buyer–property pairs, and is independently drawn from an underlying distribution. Visiting a property entails a common search cost $c > 0$.[3] Upon visiting property k, buyer i learns an idiosyncratic post-search match shock z_{ik} that is not observed prior to inspection. I assume that z_{ik} is independently drawn across buyer–property pairs from a common distribution F^z with support $[\underline{z}, \bar{z}]$, and is independent of v_{ik}. For simplicity, I assume that the transaction price is equal to the list price p_k. The utility from purchasing property k is

$$u_{ik} = v_{ik} - p_k + z_{ik}. \tag{1}$$

The information structure follows the standard Weitzman search framework. Buyers know the distribution of post-search utility before searching, and visiting a property fully reveals its realized utility. For a given buyer i, realized utilities u_{ik} are independent across properties. Conditional on property characteristics and prices, buyer-specific utility components are also independent across buyers, implying that buyers' decisions about which property they wish to purchase are independent across buyers. This independence applies to purchase intentions rather than realized transactions: multiple buyers may wish to purchase the same property, even though it can ultimately be sold to only one buyer (Albrecht et al., 2016; Ngai and Sheedy, 2024).

[2]The model abstracts from the small flows of other buyers and sellers entering the market over the relevant horizon, since the objective is to isolate the search mechanism triggered by the arrival of a new listing rather than to characterize the full dynamic evolution of the market.

[3]For simplicity, I assume that the search cost is common across buyers and properties, rather than allowing it to vary at the buyer-property level.

These assumptions imply that under optimal sequential search strategy stated in Weitzman (1979), buyer i ranks properties according to their Weitzman reservation values r_{ik},

$$r_{ik} = v_{ik} - p_k + z^*, \tag{2}$$

where z^* solves

$$\int_{z^*}^{\bar{z}} (1 - F^z(z))\, dz = c. \tag{3}$$

The distribution of reservation values is therefore induced by the distributions of prior values and property prices, rather than imposed directly. I denote this induced distribution by F^r, with support $[\underline{r}, \bar{r}]$, and assume that it admits a continuous density f^r that is bounded away from zero on its support.

For each buyer i, let $m_{ik} \in \{1, \ldots, S\}$ denote the rank of property k in the buyer's search sequence, where a lower rank corresponds to a higher reservation value. Thus, for any two properties k and j,

$$m_{ik} < m_{ij} \quad \Longleftrightarrow \quad r_{ik} > r_{ij}. \tag{4}$$

Under the optimal search rule of Weitzman (1979), buyer i visits properties in ascending order of m_{ik} (or descending order of r_{ik}) and stops searching once the highest utility realized so far, including the outside option, exceeds the reservation value of every remaining properties. Otherwise, the buyer continues searching. Let $\mathcal{N}_i$ denote the set of properties already visited. The stopping rule is given by

$$\max \left\{ u_{i0}, \max_{j \in \mathcal{N}_i} u_{ij} \right\} > \max_{k \notin \mathcal{N}_i} r_{ik}, \tag{5}$$

where u_{i0} denotes the outside option.

After stopping search, buyer i selects the property that delivers the highest realized utility among those inspected, provided that this utility exceeds her outside option u_{i0}. Following Ursu et al. (2025), I assume that every buyer makes at least one visit, so that

$$\max_{k \in \mathcal{S}} r_{ik} \geq u_{i0} \qquad \forall i. \tag{6}$$

This search rule has a central implication: properties appearing later in a buyer's search sequence are less likely to be visited and, consequently, less likely to be chosen for purchase. The intuition is that buyers may stop searching at any rank, making lower-ranked properties less likely to be visited. Moreover, these properties offer weakly lower deterministic utility, further reducing the likelihood that buyers wish to purchase them. The following lemmas formalize how visit and purchase probabilities vary with search rank.

Lemma 1 (Visiting Probability and Search Rank)**.** *Let $\nu_i(m)$ denote the probability that buyer i visits a property ranked m in her search sequence. Under the Weitzman search rule,*

$$\nu_i(m) = \begin{cases} 1, & m = 1, \\ \prod_{s=1}^{m-1} \Pr(u_{is} < r_{im}), & m \geq 2. \end{cases} \tag{7}$$

The visiting probability is strictly decreasing in search rank:

$$\nu_i(m) > \nu_i(m+1), \qquad m = 1, \ldots, S-1. \tag{8}$$

Proof. See Appendix A.1. □

Lemma 2 (Probability of Wanting to Buy and Search Rank)**.** *Let $\delta_i(m)$ denote the probability that buyer i wants to buy the property ranked m in her search sequence. Under the Weitzman search rule, this probability is strictly decreasing in search rank:*

$$\delta_i(m) > \delta_i(m+1), \qquad m = 1, \ldots, S-1. \tag{9}$$

Proof. See Appendix A.2. □

2.3 New Listing and Congestion

I now examine how new-listing entry generates congestion for incumbent sellers. A new listing may enter ahead of incumbent properties in buyers' search sequences, pushing them to lower ranks and reducing their chances of being visited and sold. Consider an incumbent property $k \in \mathcal{S}$ with rank m_{ik} for buyer i before the arrival of a new listing n. The entrant's reservation value is drawn from F^r, determining its position in the search sequence. Since reservation values are independently drawn from the same distribution, order statistics imply that the entrant is equally likely to occupy any rank in the full sequence. However, because buyers may have already visited some properties, the entrant can appear relatively early in the remaining search sequence.

Following entry, the rank of incumbent property k becomes

$$m'_{ik} = m_{ik} + \mathbf{1}\{r_{in} > r_{ik}\}. \tag{10}$$

Thus, the incumbent is pushed one position lower whenever the entrant has a higher reservation value.

Because F^r is continuous, a realized reservation value lies in the interior of its support with probability one, so any incumbent property has a strictly positive probability of being ranked both above and below a subsequently drawn listing.

$$q_{ik} \equiv \Pr(r_{in} > r_{ik}) = 1 - F^r(r_{ik}) \in (0, 1). \tag{11}$$

where q_{ik} is the ex ante probability that the entrant ranks ahead of property k. Therefore, before the entrant is realized, property k remains at rank m_{ik} with probability $1 - q_{ik}$ and is pushed to rank $m_{ik} + 1$ with probability q_{ik}.

Let $\bar{\nu}_{ik}^t$ and $\bar{\delta}_{ik}^t$ denote the expected probabilities that buyer i visits and wishes to purchase property k, respectively, where $t \in \{0, 1\}$ indicates the absence or presence of the new listing. Hence, following the arrival of the new listing, the expected visit probability from i to property k is given by:

$$\begin{aligned} \bar{\nu}_{ik}^1 &\equiv \mathbb{E}[\nu_i(m'_{ik})] \\ &= \nu_i(m_{ik}) - q_{ik}\left[\nu_i(m_{ik}) - \nu_i(m_{ik}+1)\right], \end{aligned} \tag{12}$$

and similarly, the expected probability that the buyer wishes to purchase the property is given by:

$$\begin{aligned} \bar{\delta}_{ik}^1 &\equiv \mathbb{E}[\delta_i(m'_{ik})] \\ &= \delta_i(m_{ik}) - q_{ik}\left[\delta_i(m_{ik}) - \delta_i(m_{ik}+1)\right]. \end{aligned} \tag{13}$$

Aggregating across buyers, I obtain two property-level outcomes: the expected number of buyer visits and the probability that at least one buyer wishes to purchase the property. Let V_k^t and D_k^t denote these two outcomes for property k, respectively, where $t \in \{0, 1\}$ indicates the absence or presence of the new listing.

The expected number of buyer visits is given by

$$V_k^0 = \sum_{i \in \mathcal{B}} \nu_i(m_{ik}), \qquad V_k^1 = \sum_{i \in \mathcal{B}} \bar{\nu}_{ik}^1. \tag{14}$$

Similarly, by independence across buyers, the probability that at least one buyer wishes to purchase property k is

$$D_k^0 = 1 - \prod_{i \in \mathcal{B}} \left(1 - \delta_i(m_{ik})\right), \qquad D_k^1 = 1 - \prod_{i \in \mathcal{B}} \left(1 - \bar{\delta}_{ik}^1\right). \tag{15}$$

To connect buyers' willingness to buy to the sale outcome observed in the data, suppose that property k is sold whenever at least one buyer wants to buy it.[4] If several buyers want to buy the same property, a rationing mechanism determines which buyer obtains it but does not affect whether the property is sold. Let Π_k^t denote the probability that incumbent property k is sold. Under this market-clearing assumption,

$$\Pi_k = D_k.$$

[4] If buyers anticipate that properties may be sold during their search, their optimal search strategies would need to account for the dynamic evolution of available listings. For simplicity, I adopt a static framework in which purchase and sale probabilities are determined without accounting for properties being removed from the market during the search process. This avoids the need to model optimal search strategies in a dynamic market.

The following proposition formalizes the congestion effects of new-listing entry.

Proposition 1 (Congestion from New Listings)**.** *Let Π_k^t denote the probability that incumbent property k is sold, where $t \in \{0, 1\}$ indicates the absence or presence of a new listing, respectively. The arrival of a new listing reduces both the expected number of buyer visits and the sale probability of an incumbent property:*

$$V_k^1 < V_k^0, \qquad \Pi_k^1 < \Pi_k^0. \tag{16}$$

Proof. See Appendix A.3. □

The propositions highlight a simple congestion mechanism. When a new listing enters ahead of an incumbent property, it pushes the incumbent to a lower position in buyers' search sequences. Lemmas 1 and 2 imply that lower-ranked properties are less likely to be visited and less likely to become buyers' preferred properties. New-listing entry therefore reduces both the buyer attention received by incumbent properties and their sale opportunities.

2.3.1 Price Competitiveness

The magnitude of the congestion effect depends on where the entrant appears in buyers' search sequences. Recall

$$r_{in} = v_{in} - p_n + z^*, \tag{17}$$

an entrant with a higher quality-adjusted net value, $v_{in} - p_n$, has a higher reservation value and is therefore more likely to rank ahead of incumbent properties. It consequently pushes more incumbents to lower positions in buyers' search sequences.

Proposition 2 (Stronger Congestion from More Attractive New Listings)**.** *Consider two potential entrants, n and n', such that*

$$v_{in} - p_n > v_{in'} - p_{n'} \tag{18}$$

for every buyer $i \in \mathcal{B}$. Then entrant n generates a strictly larger congestion effect on any incumbent property k:

$$V_k^1(n) < V_k^1(n'), \qquad \Pi_k^1(n) < \Pi_k^1(n'). \tag{19}$$

Proof. Since reservation values are increasing in quality-adjusted net value,

$$v_{in} - p_n > v_{in'} - p_{n'} \quad \Longrightarrow \quad r_{in} > r_{in'}.$$

Since F^r is continuous and incumbent reservation values lie in the interior of its support, it follows that

$$q_{ik}(n) = \Pr(r_{in} > r_{ik}) > \Pr(r_{in'} > r_{ik}) = q_{ik}(n').$$

Equations (12) and (13), together with Lemmas 1 and 2, imply that both the incumbent's expected visit probability and the probability that the buyer wants to buy the incumbent are strictly decreasing in q_{ik}. Aggregating across buyers therefore gives

$$V_k^1(n) < V_k^1(n'), \qquad \Pi_k^1(n) < \Pi_k^1(n').$$

□

2.4 Discussion of the Model

2.4.1 Discussion of Prices

The model holds seller prices fixed. Allowing sellers to adjust prices would make the environment dynamic and substantially more complicated, but it would also introduce an important margin of response. An incumbent seller facing greater congestion may lower the asking price, move the property forward in buyers' search sequences, and thereby recover some of the lost visits and sale opportunities. This would imply a trade-off between liquidity and price, with sellers potentially accepting lower transaction prices to mitigate congestion. I examine this mechanism empirically by testing whether new-listing entry affects list prices and transaction prices in 4.1.2.

2.4.2 Discussion of Buyer Entry

The congestion effect in the model relies on two simplifying assumptions. First, the number of buyers $\mathcal{B}$ in the local market is held fixed. Under this assumption, equations (14) and (15) characterize the decline in incumbent visits and matching opportunities following new-listing entry. In practice, however, new listings may also attract additional buyers into the local market. If buyer entry responds positively to new-listing arrivals, the larger buyer pool creates new visiting and matching opportunities and therefore works against the congestion effect generated by the reordering of existing buyers' search sequences.

To see this, let $\mathcal{B}^0$ denote the set of buyers already present in the market and let $\mathcal{E} > 0$ denote buyers who enter in response to the new listing. With buyer entry, the post-entry outcomes become

$$\widetilde{V}_k^1 = \sum_{i \in \mathcal{B}^0} \bar{\nu}_{ik}^1 + \sum_{i \in \mathcal{E}} \nu_i(m_{ik}), \tag{20}$$

and

$$\widetilde{D}_k^1 = 1 - \prod_{i \in \mathcal{B}^0} \left(1 - \bar{\delta}_{ik}^1\right) \prod_{i \in \mathcal{E}} \left(1 - \delta_i(m_{ik})\right). \tag{21}$$

Thus, the arrival of additional buyers increases both the expected number of visits and the probability that at least one buyer wishes to purchase incumbent property k, relative to the fixed-buyer case:

$$\widetilde{V}_k^1 > V_k^1, \qquad \widetilde{D}_k^1 > D_k^1. \tag{22}$$

Proposition 1 shows that new-listing entry reduces incumbent visits and purchase probabilities when the number of buyers is fixed. However, if new listings also attract additional buyers into the market, these buyers create new visit and matching opportunities for incumbent properties. This increase in buyer activity may partially or fully offset the congestion effect.

A similar point is emphasized in Anenberg and Ringo (2026), who explicitly discuss how congestion externalities can be identified in the absence of joint buyer–seller entry. My data allow me to examine this margin directly by testing whether new-listing entry attracts additional buyers into the local market. I return to this issue in Section 4.4.

3 Data and Empirical Strategy

3.1 Housing Dataset

The dataset is from China's largest multiple listing service (MLS), which covers buyer visits to sellers, seller listings and listing price revisions, as well as transaction data from Beijing during August 2015 to July 2016. The agency company was established in Beijing and holds more than 50% of the market share in Beijing, making this dataset highly representative. The data provide detailed information on buyers' search histories, particularly records of in-person visits to individual properties. They also cover property listings and transactions on the platform, including rich information on property characteristics, locations, listing and transaction prices, and the complete history of listing price adjustments.

To study how new listings generate congestion for existing properties, I first distinguish between new and existing listings. To do so, I divide the one-year sample period into two six-month windows. Properties listed during the first six months and still on the market at the beginning of the second period form the stock of *existing listings*, while properties entering during the second period are defined as *new listings*. I use the second six-month period as the estimation window and ex-

clude new listings themselves from the estimation sample, as the analysis focuses on their effects on incumbent properties.

3.2 Identification Strategy

The identification strategy closely follows Anenberg and Ringo (2026), who exploit the arrival of an individual new listing as a local supply shock and examine how nearby transaction volume and prices respond following its arrival. Similarly, I exploit variation in the timing of new-listing arrivals to identify their effects on properties already on the market.

My baseline design defines exposure at the building level. A building is treated when it experiences exactly one new-listing arrival during the observation period, while buildings with no new-listing arrivals serve as controls; buildings with multiple arrivals are excluded. This narrow market definition is useful both economically and for identification. Properties within the same building share highly similar location and structural characteristics and are therefore likely to be close substitutes for prospective buyers. At the same time, restricting exposure to the building level avoids the repeated and overlapping treatments that would arise under broader geographic definitions.

Identification comes from comparing changes in incumbent outcomes across nearby buildings that experience new-listing arrivals at different times. My preferred specification includes apartment-complex-by-week fixed effects, which absorb common time-varying conditions within the same local housing market. The identifying assumption is that, absent new-listing entry, incumbent properties in treated and control buildings would have followed parallel short-run trends in buyer visits and sale probabilities. This assumption does not require listing decisions or their timing to be random. The remaining concern is a building-specific time-varying shock that both induces a new listing and independently affects incumbent outcomes. I assess this possibility using event-study specifications and examine whether treated and control properties display differential trends prior to entry.

I complement the baseline design with two alternative specifications. First, I implement a apartment-complex-level design to study transaction prices and to test whether congestion extends beyond the same building.[5] At this level, a treat-

[5]In China, a apartment complex is typically a gated community consisting of several apartment buildings developed by the same developer and constructed around the same time. Properties within a complex share the same location and neighborhood amenities and are highly homogeneous in construction quality and architectural characteristics. Prices are therefore highly comparable within a complex, with most remaining differences arising from unit-level characteristics such as floor area, number of rooms, and floor level.

ment event is defined as a week in which at least M new listings enter a complex. I retain complexes that experience exactly one such event and complexes that never experience one, while excluding complexes with multiple treatment events. I vary M across specifications to capture different intensities of new-listing entry. The results are presented in 4.3.1.

Second, I use the full sample of buildings, including those with multiple new-listing arrivals. In this specification, I use the cumulative number of new listings as a time-varying treatment measure in a TWFE framework and use the timing of the first new-listing arrival to define treatment timing in an event study. I treat this design as a robustness exercise rather than the baseline because repeated treatments may contaminate subsequent treatment effects, and many buildings experience their first arrival near the beginning of the observation window, leaving limited pre-treatment history. The results are presented in 4.3.2.

3.3 Model Specification

I start the estimation with a canonical two-way fixed effect specification:

$$Y_{ibt} = \alpha + \beta NewListing_{bt} + X_i'\eta + \theta_{it} + \gamma_b + \rho_n\phi_t + \epsilon_{ibt}, \tag{23}$$

Y_{ibt} denotes the outcome of existing property i in building b in week t, including the number of buyer visits, the probability of sale, prices, and other seller outcomes. $NewListing_{bt}$ is an indicator equal to one once property i is exposed to a new-listing shock in building b at week t. X_i is a vector of time-invariant housing characteristics, including log floor area, floor level, and the numbers of bedrooms, bathrooms, and living rooms.[6] θ_{it} denotes time-on-market fixed effects, which flexibly control for duration dependence in property outcomes by absorbing differences associated with the number of weeks property i has been on the market. Controlling for time on market is particularly important because buyer attention and sale outcomes exhibit strong duration dependence: properties that remain unsold for longer periods may systematically receive fewer visits and have lower probabilities of sale.

γ_b denotes building fixed effects, which absorb time-invariant differences across buildings.[7] Finally, $\rho_n \times \phi_t$ denotes complex-by-week or neighborhood-by-week

[6] Within the same building, these are the only characteristics that retain variation; other attributes, such as building age, amenities, and building height, are absorbed by building fixed effects.

[7] I include building fixed effects together with observable property characteristics rather than property fixed effects because my baseline specification is intended to capture changes in outcomes among the set of existing listings within a building following the arrival of a new listing. As prop-

fixed effects,[8] which flexibly absorb time-varying local market conditions common to properties within the same complex or neighborhood. My preferred specification includes complex-by-week fixed effects, thereby identifying the effect of a new-listing shock from variation across buildings within the same apartment complex and week.

Causal identification relies on the parallel trends assumption: absent new-listing entry, outcomes for treated and control properties would have evolved similarly. I assess this assumption using an event-study specification that traces buyer visits and sale probabilities around new-listing entry. The absence of differential pre-trends would support this assumption and help rule out endogenous entry in response to changes in incumbent outcomes. Specifically, I estimate the following specification, where T_b denotes the week in which building b receives a new listing. The week immediately preceding entry ($\tau = -1$) serves as the omitted period, and event times outside the $[-7, 7]$ window are binned at the respective endpoints.

$$Y_{ibt} = \alpha + \sum_{\tau=-7, \tau\neq-1}^{7} \beta_\tau \mathbb{I}(t - T_b = \tau) + X_i'\eta + \theta_{it} + \gamma_b + \rho_n\phi_t + \epsilon_{ibt}. \tag{24}$$

Recent advances in the DiD literature have shown that, under staggered treatment adoption, conventional two-way fixed effects (TWFE) estimators may be biased when treatment effects vary across cohorts or over time (Callaway and Sant'Anna, 2021; de Chaisemartin and D'Haultfœuille, 2020; Goodman-Bacon, 2021). In particular, TWFE estimates can combine heterogeneous treatment effects using potentially negative weights, which may distort the estimated average treatment effect. To address this concern, I report the conventional TWFE estimates together with the interaction-weighted estimator proposed by Sun and Abraham (2021). This estimator constructs cohort-specific treatment effects using appropriate comparison groups and then aggregates them, thereby avoiding the problematic comparisons and negative weighting that can arise under conventional TWFE estima-

erties are sold and exit the market, the composition of the at-risk listings naturally evolves over time. Property fixed effects would place greater weight on within-property changes for listings that remain observed over time, thereby abstracting from part of this evolution in the risk set. This is particularly restrictive for the sale outcome, which is defined as an indicator equal to 100 in the week a property sells and zero in all other observed weeks. With property fixed effects, identification would therefore rely on highly sparse within-property variation concentrated in a single sale week. By contrast, building fixed effects allow the composition of existing listings to change before and after treatment, while observable property characteristics account for differences in the composition of listings and building fixed effects absorb persistent heterogeneity across buildings.

[8] A *neighborhood* is a local housing-market area defined by the real estate platform rather than an administrative unit. The platform divides Beijing into slightly more than 200 such areas. Although relatively broad, the neighborhood is the next geographic level above the apartment complex in the platform's market hierarchy.

tion.

3.4 Summary Statistics

I first report descriptive patterns in the housing market over the sample period. From August 2015 to July 2016, both transaction prices and list prices per square meter increased steadily (see Figure A1). In contrast, the numbers of new listings and transactions remained relatively stable over time, apart from the sharp decline around the Chinese New Year. The number of new listings consistently exceeded the number of transactions, suggesting a gradual accumulation of for-sale inventory over the sample period.[9]

Figure A2 further shows the evolution of aggregate buyer visits and market tightness. The total number of buyer visits remained relatively stable over time, again with the exception of the Chinese New Year period. Market tightness, measured as the ratio of total buyer visits to the number of active listings, declined over the sample period. This decline is consistent with the expansion of for-sale inventory relative to a relatively stable level of buyer attention. From the seller's perspective, this implies increasing competition for a limited pool of buyer attention, providing a useful setting in which to study congestion effects.

I construct the main estimation sample as a property-week panel, with summary statistics for the main variables reported in Table 1. Panel A presents property-level characteristics, including housing attributes, the initial list price, and transaction outcomes. Panel B reports variables measured at the property-week level, including weekly buyer visits, list prices, the sale indicator, and the new-listing treatment indicator. I also report property characteristics separately for the treatment and control groups in Table A1. Although t-tests reject equality of means for several characteristics, the standardized differences are small, suggesting that the two groups are broadly similar in observable housing attributes. In particular, the average initial list price differs by only about 2.1% between the two groups, indicating a relatively small difference in pricing.

I also report summary statistics for the full building-level sample in Table A2, alongside those for the apartment-complex-level samples under alternative treatment thresholds in Table A3. I use these samples for alternative estimations, particularly for examining transaction prices, as the baseline sample contains relatively few transactions.

[9]One caveat is that, due to data limitations, I cannot fully observe properties that exit the market without a transaction. The measured stock of active listings may therefore be overstated, which may mechanically contribute to the observed increase in inventory and decline in measured market tightness.

Table 1: Summary Statistics for Baseline Sample

Variables	N	Mean	Std. Dev.	Min	Max
Panel A: Property-Level Characteristics					
Total Visits	18,237	3.158	6.670	0	38
Sale Indicator	18,237	14.811	35.521	0	100
Log Initial List Price	18,237	14.882	0.559	13.122	16.787
Living Rooms	18,237	1.274	0.521	0	2
Bedrooms	18,237	2.397	0.807	1	5
Bathrooms	18,237	1.319	0.546	1	3
Log Floor Area	18,237	4.485	0.434	3.535	5.680
Floor Level	18,237	4.567	3.413	1	27
Panel B: Property-Week Outcomes					
Visits	487,200	0.125	0.713	0	29
Sale Indicator	487,200	0.554	7.425	0	100
Log List Price	487,200	14.885	0.573	13.122	16.760
New Listing	487,200	0.300	0.458	0	1

Notes: Panel A reports summary statistics at the property level, while Panel B reports summary statistics for the property-week panel used in the baseline analysis. Total Visits is the total number of buyer visits received by a property. Sale Indicator equals 100 if the property is sold and zero otherwise. Sale Indicator in Panel B equals 100 in the week in which a property is sold and zero otherwise. New Listing is an indicator equal to one after the arrival of a new listing in the same building.

4 Empirical Results

4.1 Main Results of Congestion Effects

4.1.1 Visits and Sale Probability

I first test the main prediction of Proposition 1: the arrival of a new listing should reduce both the number of visits received by incumbent properties and their probability of sale. Table 2 reports the estimates under different specifications. Columns (1)–(3) examine the effect on the number of visits. Across all specifications, the coefficient on New Listing is negative and statistically significant at the 1% level. The magnitude remains similar after progressively absorbing local time-varying shocks using neighborhood-by-week and complex-by-week fixed effects. The estimated effect ranges from -0.031 to -0.037 visits per week using the TWFE estimators, and from -0.048 to -0.052 using the Sun and Abraham estimators. Relative to the pre-treatment average of approximately 0.21 weekly visits among treated

properties, these estimates correspond to an overall decline of approximately 15–25% in buyer visits following the arrival of a new listing in the same building. Figure 1 presents the corresponding event-study estimates. Before the new listing arrives, the treatment and control groups exhibit similar trends in weekly visits, providing support for the parallel-trends assumption. Following the arrival of the new listing, visits to existing properties decline after 1 week and remain below their pre-treatment level. The dynamic pattern therefore suggests that new listings divert buyer attention away from existing properties in the same building.

Columns (4)–(6) of Table 2 examine whether this loss of buyer visits translates into a lower probability of sale. The estimated coefficients are consistently negative, ranging from -0.106 to -0.127 percentage points using the TWFE estimators and from -0.187 to -0.252 percentage points using the Sun and Abraham estimators. Relative to an average weekly sale probability of 0.93 p.p. among treated properties before treatment, these estimates correspond to an overall decline of approximately 11–27% in the weekly probability of sale. Figure 2 shows a similar dynamic pattern: sale probabilities exhibit no differential trends before treatment, begin to decline following the arrival of a new listing, and become significantly negative four weeks after the shock.

To further examine the role of buyer visits, I re-estimate the sale-probability regressions while controlling for the number of weekly visits received by each incumbent property. These estimates can be interpreted as the effect of new-listing entry on sale probability conditional on buyer visits. Table A4 shows that the estimated congestion effects become smaller in magnitude and generally less statistically significant after controlling for visits. Most coefficients are no longer statistically significant, while the remaining estimates are only weakly significant. This attenuation suggests that reduced buyer visits are an important channel through which new-listing entry lowers incumbent sale probabilities, consistent with the model's search-ranking mechanism.

Taken together, these results point to a substantial congestion externality operating primarily through buyer visits. The similar percentage declines in visits and sale probabilities, together with the attenuation of the congestion coefficients after controlling for visits, suggest that reduced buyer attention accounts for an important part of the decline in incumbent sales. The latter margin would be more consistent with changes in unobserved property quality or selection among the listings that remain on the market. Instead, the results indicate that congestion arises mainly because new listings alter buyers' search rankings and divert visits away from incumbent properties.

Table 2: The Impact of New Listings on Visits and Sale Probability

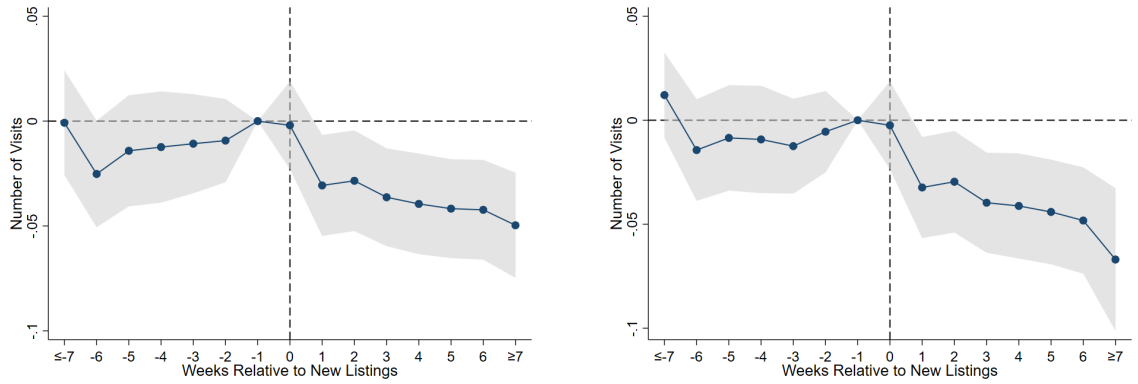
	Number of Visits			Sale Probability		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: TWFE Estimators						
New Listing	-0.037***	-0.036***	-0.031***	-0.121***	-0.106***	-0.127***
	(0.006)	(0.006)	(0.008)	(0.039)	(0.039)	(0.046)
Panel B: Sun and Abraham (2021) Estimators						
New Listing	-0.052***	-0.051***	-0.048***	-0.194**	-0.187**	-0.252***
	(0.010)	(0.010)	(0.012)	(0.079)	(0.080)	(0.092)
N	487,200	487,170	459,273	487,200	487,170	459,273
Housing Attributes	✓	✓	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighbor × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: Panel A reports estimates from the two-way fixed effects specification. Panel B reports estimates using the interaction-weighted estimator of Sun and Abraham (2021), which accommodates treatment-effect heterogeneity across treatment cohorts and event time. The dependent variables are the number of buyer visits received by a property in a given week and an indicator for whether the property is sold in that week. Housing controls include floor level, floor area, and the numbers of kitchens, bedrooms, and living rooms. Standard errors are clustered at the building level. *** p<0.01, ** p<0.05, * p<0.1.

4.1.2 Prices

Incumbent sellers may respond to new-listing congestion through the price margin. When a new listing enters ahead of an incumbent property in buyers' search sequences, the incumbent becomes less likely to be visited and sold. By lowering its asking price, the incumbent raises its reservation value from the buyer's perspective and moves back toward the front of the search sequence. Price adjustment therefore allows sellers to partially offset the congestion effect by improving liquidity, but only at the cost of accepting a lower sale price. The model thus implies a trade-off between price and liquidity: stronger congestion may induce incumbent sellers to sacrifice price in order to recover buyer visits and sale opportunities.

I therefore examine whether new-listing entry affects transaction prices. Because transaction prices are observed only for properties that sell, the number of transactions within narrowly defined buildings is relatively small. I therefore conduct the main transaction-price analysis at the apartment-complex level as described in 3.2. Specifically, I define a complex as treated if it experiences exactly



(a) TWFE Estimators

(b) Sun and Abraham Estimators

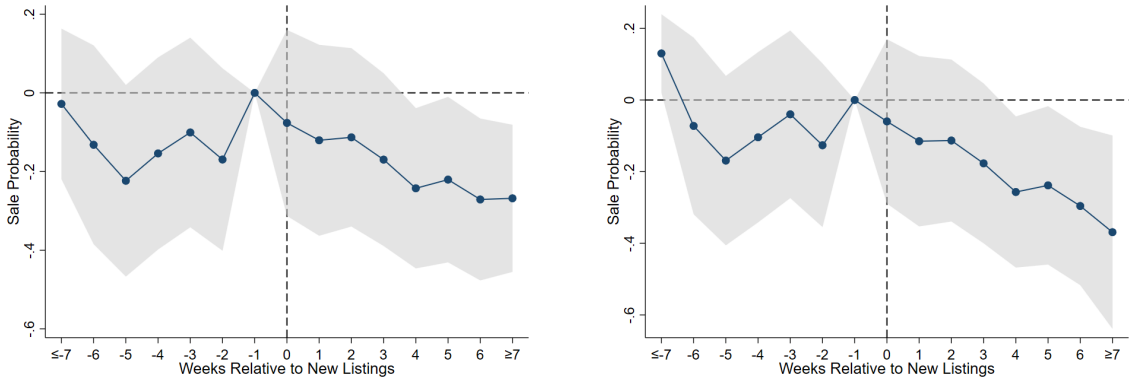
Figure 1: The Dynamic Effects of New Listings on Visits

Notes: The figure shows the dynamic treatment effects of a new-listing shock at the building level on the number of visits. Panel (a) reports estimates from the two-way fixed effects (TWFE) event-study specification, while Panel (b) reports estimates using the Sun and Abraham estimator. Both panels correspond to the specification reported in Column (3) of Table 2. The shaded areas represent 95% confidence intervals.

one treatment event during the sample period, where a treatment event occurs when at least M new listings enter the complex within a week. Complexes with fewer than M new listings in every week throughout the observation period form the control group. Complexes that experience more than one such treatment event are excluded. I vary the threshold M across specifications to examine whether transaction prices respond to different intensities of new-listing entry.

Table 3 reports the results, and Figure A3 presents the corresponding event-study estimates. Across specifications, I find no significant effect of new-listing arrivals on transaction prices except in column (5). However, the corresponding event-study estimates for column (5) in Figure A3 exhibit significant pre-treatment differences, casting doubt on a causal interpretation of this coefficient. Taken together, there is no strong evidence suggesting that incumbent sellers systematically respond to new-listing congestion by accepting lower sale prices. In Section 4.3.1, I further examine the effect of cumulative new listings on transaction prices using the full building-level panel and again find no significant effects. I also examine list-price responses at the building level using the baseline treatment definition. As shown in Figure 3, I find no evidence that incumbent sellers adjust their asking prices following the arrival of a new listing.

Despite substantial declines in buyer visits and sale probabilities, new-listing does not appear to induce meaningful price adjustments among incumbent properties. This finding is also consistent with Anenberg and Ringo (2026), who show that, after controlling for housing characteristics, both transaction prices and the ratio of transaction prices to list prices are largely unaffected by the arrival of new



(a) TWFE Estimators

(b) Sun and Abraham Estimators

Figure 2: The Dynamic Effects of New Listings on Sales

Notes: The figure shows the dynamic treatment effects of a new-listing shock at the building level on the sale probability. Panel (a) reports estimates from the two-way fixed effects (TWFE) event-study specification, while Panel (b) reports estimates using the Sun and Abraham estimator. Both panels correspond to the specification reported in Column (6) of Table 2. The shaded areas represent 95% confidence intervals.

listings. Overall, the evidence suggests that the congestion effect of new listings operates primarily through the liquidity margin rather than the price margin. Incumbent sellers therefore appear to absorb increased competition mainly through a longer expected time to sale rather than through lower prices.

4.2 Heterogeneity in Congestion Effects

I next examine heterogeneity in the congestion effect along four dimensions. First, I examine the price competitiveness of entrants to directly test a prediction of the sequential-search model. I then consider incumbent listing age, property substitutability, and local market size to further characterize the mechanisms and conditions under which congestion is more pronounced.

4.2.1 Price Competitiveness of New Listings

Proposition 2 predicts stronger congestion from entrants with higher quality-adjusted net value. I test this prediction by comparing new listings whose price per square meter is lower versus higher than that of the incumbent property.[10] A lower price per square meter makes the entrant more price competitive and more likely to rank ahead of the incumbent in buyers' search sequences. Because market prices rose

[10]Because the new and existing listings are located in the same building, they share nearly identical housing and location characteristics. Price per square meter therefore provides a simple measure of relative price competitiveness while accounting for differences in unit size.

Table 3: The Impact of New Listings on Sale Prices

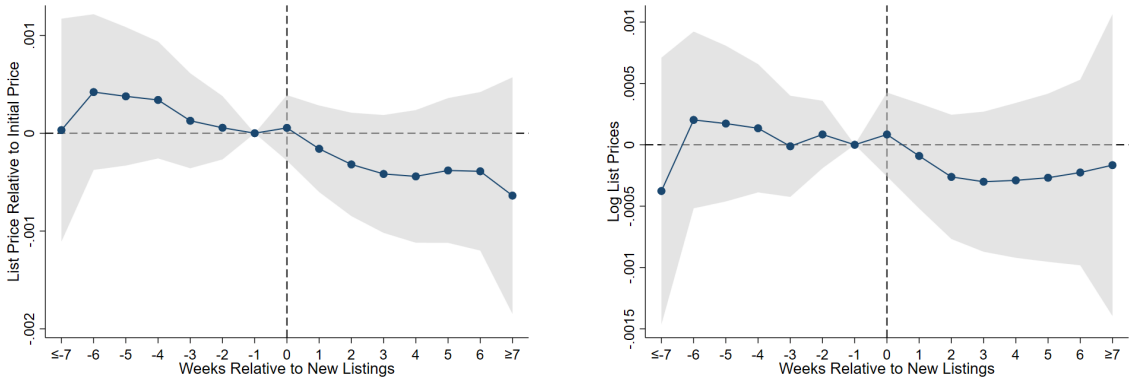
$M =$	Treatment: At Least M New Listings at Complex-Week					
	1	2	3	4	5	6
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	0.092**	0.006	0.001	-0.001	-0.008**	-0.003
	(0.036)	(0.007)	(0.004)	(0.004)	(0.004)	(0.004)
N	95	3,276	10,410	18,800	26,815	33,542
R^2	0.991	0.984	0.979	0.975	0.971	0.969
Housing Attributes	✓	✓	✓	✓	✓	✓
Complex F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the apartment complex level. This table reports the effects of new listings on sale prices at the apartment complex level. In column M, a complex-week is classified as treated if at least N new properties are listed within the complex during that week, while complex-weeks with fewer than M new listings serve as the control group. To preserve a sufficiently large sample of transactions and avoid dropping numerous singleton observations, I do not include high-dimensional fixed effects in these specifications.

continuously during the sample period, newly listed properties are typically more expensive than existing listings, as shown in Figure A1. Figure A4 shows that only about 13% of new listings are priced below the corresponding incumbent property, while some are priced as much as 70% higher.

Table 4 examines heterogeneity by the price competitiveness of the new listing. I split treated buildings according to whether the entrant is priced below or above the incumbent property and estimate the two specifications separately, using the same never-treated buildings as the control group in both regressions. Consistent with Proposition 2, more price-competitive entrants generate substantially stronger congestion effects. A new listing priced below the incumbent property reduces weekly visits by about 0.13 and the weekly probability of sale by 0.88 percentage points, whereas a more expensive entrant reduces weekly visits by only 0.03 and the sale probability by 0.19 percentage points.[11] Tests of equality reject that the treatment effects are the same across the two groups for both outcomes.

[11] Because these estimates come from separate regressions with different treated samples, while sharing the same control group, the two coefficients need not mechanically aggregate to the baseline treatment effect. In particular, the fixed effects and covariates are re-estimated within each subsample, so the identifying variation and implicit regression weights differ across the split-sample specifications and the pooled baseline regression.



(a) TWFE Estimators

(b) Sun and Abraham Estimators

Figure 3: The Dynamic Effects of New Listings on List Prices

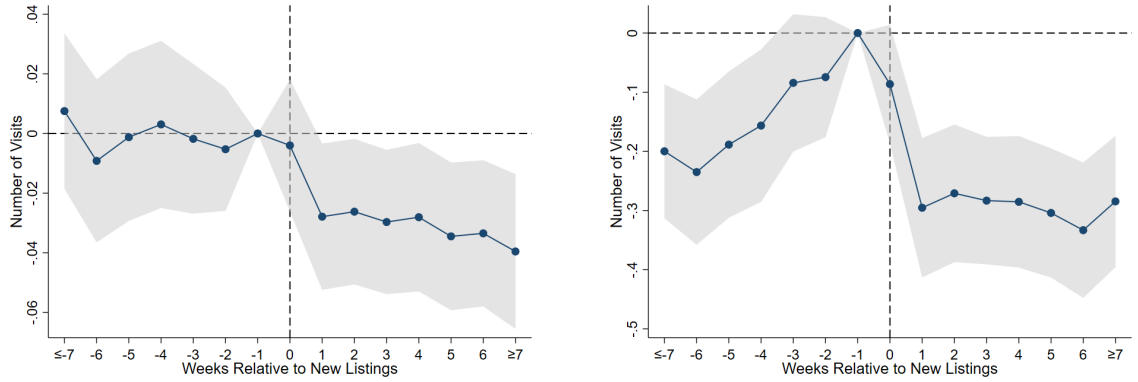
Notes: The figure shows the dynamic treatment effects of new listings shock within the same building on list prices of the existing sellers. The shaded areas represent 95% confidence intervals.

The effects of cheaper entrants are therefore approximately four times as large as those of more expensive entrants.

Figure 4 presents the corresponding event-study estimates separately for cheaper and more expensive new listings. For cheaper new listings, the pre-treatment coefficients exhibit some upward movement, which warrants caution in interpreting the estimates. Importantly, however, the arrival of a cheaper new listing coincides with a sharp reversal of the pre-existing upward pattern in both buyer visits and sale probability, producing a clear discontinuous decline around the treatment date. I therefore interpret the event-study evidence as suggestive that the substantially larger effects of cheaper new listings are not simply driven by continuation of the pre-treatment trend.

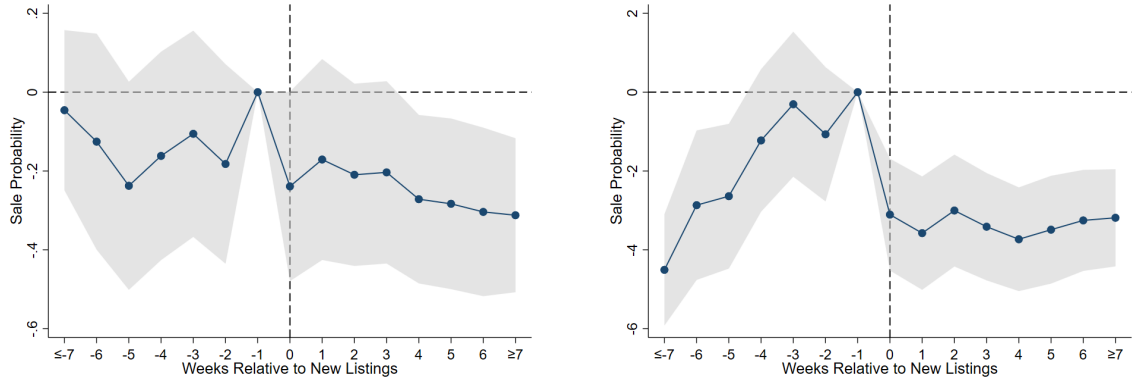
4.2.2 Listing Age of Incumbent Sellers

The sequential-search model describes how buyers match with sellers through an ordered search process. Buyer–seller matching may also exhibit a stock–flow structure, whereby newly listed properties initially match with the stock of buyers already active in the market, while listings that remain unsold for longer increasingly rely on the flow of newly arriving buyers (Smith et al., 2026). I provide direct descriptive evidence of this pattern using the observed search histories of buyers and sellers. Figure A5 plots the average and median time on market of buyers visiting a property against the property's time on market. The figure reveals a clear negative relationship: recently listed properties tend to attract buyers who have been searching for longer, whereas older listings are increasingly visited by buyers who have only recently entered the market. This pattern is consistent with stock–flow matching, in which new listings initially encounter the existing stock of buyers,



(a) Expensive New Listings and Visits

(b) Cheap New Listings and Visits



(c) Expensive New Listings and Sales

(d) Cheap New Listings and Sales

Figure 4: The Dynamic Effects of Heterogeneous Price Competitive New Listings

Notes: The figure shows how number of visits of the existing listings changes after new listings shock at the building level. The panel (a) shows the dynamics when houses are affected by more expensive new listing shocks, while the panel (b) illustrates the dynamics when houses are affected by cheaper new listing shocks. The shaded areas represent 95% confidence intervals.

while older listings increasingly depend on the flow of new buyers.

This stock–flow structure yields an implication for the incidence of congestion across listing ages. Recently listed incumbents still compete with new listings for the existing stock of buyers and are therefore more exposed to congestion. By contrast, older incumbents have exhausted many of these initial matching opportunities and rely more heavily on newly arriving buyers. Congestion should therefore be concentrated among relatively recent incumbents. To examine this heterogeneity, I divide treated incumbent properties according to their time on market when the new listing arrives. I use the median time on market in the treated sample, 32 days, as the cutoff. Properties listed for no more than 32 days at the time of treatment are classified as *Recent*, while those listed for more than 32 days are classified as *Old*. I then estimate the baseline specification separately for the two treated subsamples, using the same never-treated properties as the control group in both regressions.

Table 4: The Price Competitiveness of New Listings

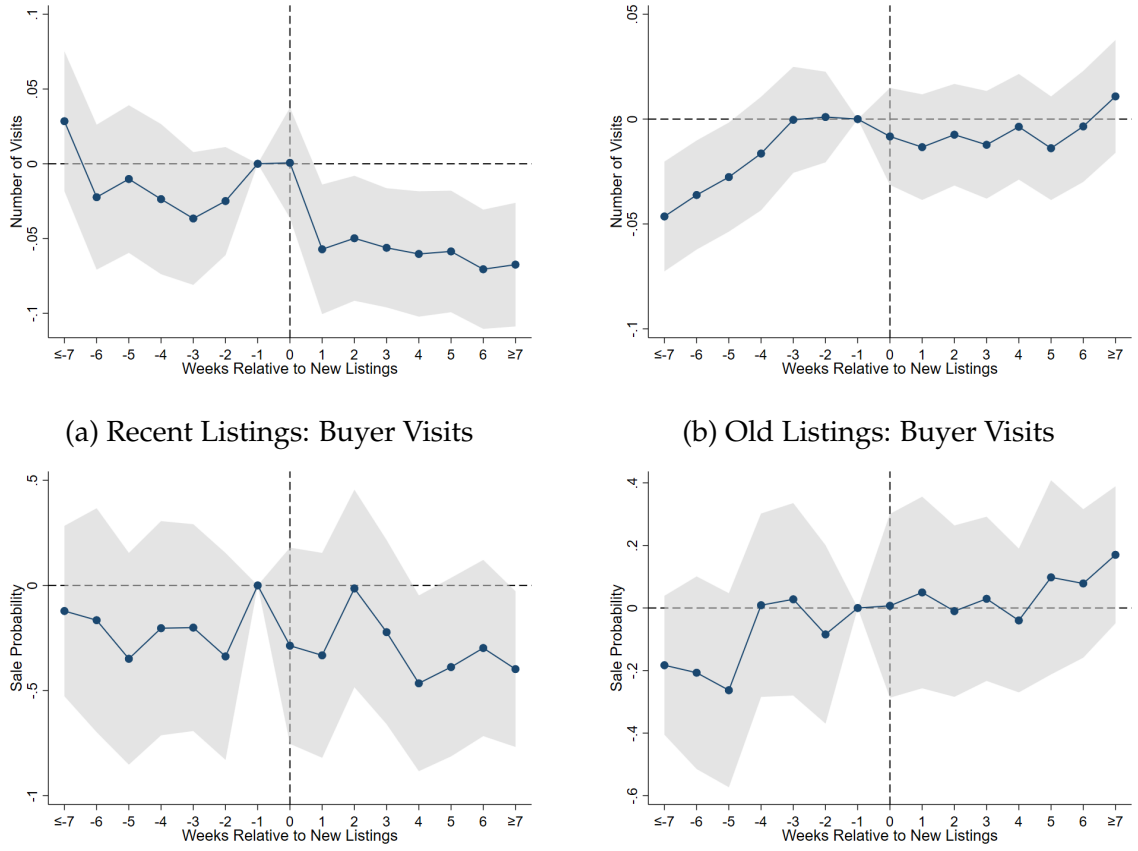
	Number of Visits		Sale Probability	
	Cheap	*Expensive*	*Cheap*	*Expensive*
	(1)	(2)	(3)	(4)
New Listing	-0.130***	-0.031***	-0.882***	-0.193***
	(0.028)	(0.008)	(0.176)	(0.047)
N	230,093	425,531	230,093	425,531
R^2	0.453	0.347	0.341	0.253
Housing Attributes	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓
Complex × Week F.E.	✓	✓	✓	✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. New listings are classified as *Cheap* (*Expensive*) if their price per square meter is lower (higher) than that of the corresponding existing listing. For the *Cheap* specifications, the treatment group consists of existing listings exposed to a cheaper new listing and is compared with the control group. For the *Expensive* specifications, the treatment group consists of existing listings exposed to a more expensive new listing and is compared with the same control group.

Table 5 reports the results and Figure 5 presents the corresponding event-study plots. The negative congestion effect is concentrated among recently listed properties. For the *Recent* group, the arrival of a new listing reduces weekly buyer visits by 0.054 and the weekly sale probability by 0.155 percentage points. In contrast, the coefficients for the *Old* group are positive. These estimates should not be interpreted as evidence that new listings benefit older incumbents, however, because Figure 5 shows differential pre-treatment trends for this group, making a causal interpretation difficult. Overall, the concentration of the negative effect among recent listings is consistent with the stock–flow matching mechanism, under which recent incumbents compete more directly with new listings for the existing stock of active buyers.

4.2.3 Substitutability of New Listings

I next examine whether the congestion effect depends on the substitutability between the new listing and existing properties. If the negative externality operates through the diversion of buyer attention, its magnitude should depend on



(a) Recent Listings: Buyer Visits

(b) Old Listings: Buyer Visits

(c) Recent Listings: Sale Probability

(d) Old Listings: Sale Probability

Figure 5: Dynamic Effects of New Listings by Listing Age of Incumbent Sellers

Notes: This figure presents event-study estimates of the effects of new-listing arrivals on incumbent properties, separately by their time on market at the treatment date. *Recent* listings are properties that have been on the market for no more than 32 days when the new listing arrives, while *Old* listings have been on the market for more than 32 days. Panels (a) and (b) report the effects on buyer visits, and Panels (c) and (d) report the effects on sale probability. The shaded areas represent 95% confidence intervals.

how closely the new listing competes with an existing property for the same buyers. A new listing that is similar to an existing property provides a closer alternative and should therefore divert more buyer attention from that property. I measure substitutability along two salient housing characteristics—floor area and floor level—which are the only housing attributes that vary within buildings. I split the treated sample by whether the new and incumbent listings have the same floor area or floor level, and estimate each subgroup separately using the same control group. I expect the negative effect of new listings to be stronger when the two properties share these characteristics.

Table 6 reports the results. The congestion effect on buyer visits is substantially stronger when new and incumbent properties share similar characteristics. In particular, the decline in visits is approximately four times larger when the two properties have the same floor area or are located on the same floor. The results for

Table 5: Heterogeneity by Listing Age of Incumbent Sellers

	Number of Visits		Sale Probability	
	Recent	*Old*	*Recent*	*Old*
	(1)	(2)	(3)	(4)
New Listing	-0.054***	0.025***	-0.155**	0.197***
	(0.013)	(0.008)	(0.068)	(0.046)
N	323,020	313,855	323,020	313,855
R^2	0.374	0.363	0.279	0.280
Housing Attributes	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓
Complex × Week F.E.	✓	✓	✓	✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. Treated incumbent properties are classified according to their time on market when the new listing arrives. *Recent* properties are those that have been listed for no more than 32 days, the median time on market in the treated sample, while *Old* properties are those that have been listed for more than 32 days. Each treatment group is compared with the control group of properties in buildings that experience no new-listing arrival during the sample period.

sale probability generally follow a similar pattern, with stronger negative effects for closer substitutes, although the estimate for properties with the same floor area is imprecise and statistically insignificant. Overall, these findings suggest that new listings divert more buyer search from incumbent properties when they are closer substitutes, highlighting the role of property substitutability in shaping the incidence of congestion.

4.2.4 Market Size

I next examine whether the congestion effect varies with the size of the local market. Market size is important because the competitive impact of a new listing should depend not only on its arrival but also on the number of properties already available to buyers. In a smaller market, the addition of a single listing represents a larger proportional expansion of the relevant choice set and may therefore attract a larger share of buyer attention away from existing properties. By contrast, in a larger market with many competing listings, one additional property constitutes a relatively small change in the available inventory and should have a more limited

Table 6: The Substitutability of New Listings

	Number of Visits		Sale Probability	
	Same	*Different*	*Same*	*Different*
	(1)	(2)	(3)	(4)
Panel A: Floor Area				
New Listing	-0.099***	-0.023***	-0.319***	-0.117**
	(0.023)	(0.008)	(0.113)	(0.050)
N	216,957	428,705	216,957	428,705
R^2	0.413	0.337	0.320	0.246
Panel B: Floor Level				
New Listing	-0.090***	-0.020**	-0.412***	-0.117**
	(0.022)	(0.008)	(0.124)	(0.049)
N	220,603	424,040	220,603	424,040
R^2	0.416	0.334	0.320	0.247
Housing Attributes	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓
Complex $\times$ Week F.E.	✓	✓	✓	✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. Panel A classifies treated listings according to whether the new listing has the same floor area as the corresponding existing listing. The *Same* (*Different*) specifications compare existing listings exposed to a new listing with the same (different) floor area with the control group. Panel B follows the same procedure based on floor level: the *Same* (*Different*) specifications compare existing listings exposed to a new listing on the same (different) floor with the control group.

Table 7: Market Size and Congestion Effects of New Listings

	Number of Visits			Sale Probability		
	(1)	(2)	(3)	(4)	(5)	(6)
New Listing× Market Size	0.009*	0.007*	0.013**	0.086***	0.071***	0.079***
	(0.005)	(0.004)	(0.005)	(0.020)	(0.020)	(0.025)
New Listing	-0.056***	-0.051***	-0.057***	-0.267***	-0.215***	-0.242***
	(0.013)	(0.012)	(0.014)	(0.068)	(0.068)	(0.079)
N	484,819	484,789	456,944	484,819	484,789	456,944
R^2	0.184	0.199	0.314	0.070	0.084	0.220
Housing Attributes	✓	✓	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighbor × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. This table examines whether the effect of new listings varies with local market size. Market Size is measured by the number of existing listings in the building at the time the new listing enters the market. The interaction term, New Listing × Market Size, therefore captures how the effect of a new-listing arrival varies with the number of incumbent properties competing in the same building. A positive interaction coefficient indicates that the negative effect of a new listing becomes weaker as market size increases.

effect on the allocation of buyer attention. I therefore expect the negative effect of new listings on existing properties to be stronger in smaller markets.

I measure market size by the number of existing listings in the building at the time a new listing enters the market and interact this measure with the new-listing indicator. The results are reported in Table 7. The interaction coefficients are consistently positive, indicating that the negative effects of new listings become weaker as market size increases. This pattern is consistent with the congestion mechanism: in a larger market, the arrival of one additional listing represents a smaller proportional expansion of the set of available properties and therefore diverts less buyer attention from each existing listing.

4.3 Alternative Specifications

4.3.1 Multiple New Listings

The baseline analysis restricts the sample to buildings that experience either one or no new-listing arrival during the sample period. While this restriction provides a clean comparison that isolates the effect of a single new listing, it may introduce a sample-selection concern by excluding buildings with more active listing

entry. To assess whether my findings depend on this restriction, I retain the full sample and allow buildings to experience multiple new-listing arrivals. For each building-week, I construct the cumulative number of new listings that have entered the building up to that week and regress incumbent properties' buyer visits, sale probability, and prices on this measure. This specification allows me to examine whether the baseline congestion pattern extends to the broader sample and to settings with repeated new-listing entry.

Tables A5 and A6 present the results. Table A5 shows that the cumulative number of new listings is significantly negatively associated with both buyer visits and the sale probability of existing listings. Each additional new listing is associated with 0.001–0.004 fewer weekly buyer visits and a 0.018–0.026 percentage-point decline in the weekly probability of sale, corresponding to approximately 0.7–2.8% and 3.3–4.8% of their respective sample means (see Table A2). Given an average of 4.278 cumulative new listings, these estimates imply reductions of approximately 3–12% in weekly visits and 14–20% in weekly sale probability relative to their sample means. These implied effects are somewhat smaller than, but quantitatively comparable to, the baseline estimates. Table A6 shows no significant association with either list or sale prices. Overall, these results reinforce the baseline finding that new-listing entry primarily reduces incumbent liquidity rather than prices, even when buildings experiencing multiple new-listing arrivals are included.

Using the same full panel, I also construct an event study in which treatment timing is defined by the first new-listing arrival in each building during the sample period. Figure A7 presents the results. Although the estimates show some evidence of differential pre-treatment trends, the declines in buyer visits and sale probabilities occur immediately and become clearly visible following the first new-listing arrival. The similarity of these patterns to the baseline estimates suggests that the sample restriction to buildings with a single new-listing arrival does not materially drive the main results.

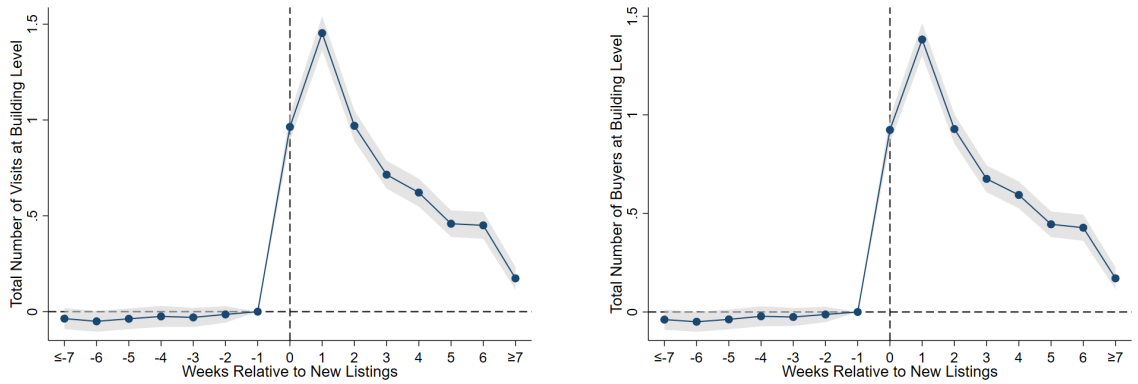
4.3.2 Apartment-Complex-Level Evidence

To examine the geographic scope of congestion and the effects of simultaneous new-listing arrivals, I extend the analysis to the apartment-complex level. Applying the baseline treatment definition at this broader spatial level leaves relatively few eligible complexes, because new listings arrive frequently and many complexes experience multiple treatment events during the sample period. I therefore modify the treatment definition. Specifically, I define a complex as treated if it experiences at least M new listings within a week, while complexes with fewer than M new listings in every week throughout the observation period form the control

group. I vary the threshold N across specifications to obtain sufficient treated and control observations and to examine how the congestion effect changes with the intensity of new-listing entry.

Tables A7 and A8 report the estimation results, and Figures A8 and A9 present the corresponding event-study estimates. I find that new-listing entry significantly reduces both buyer visits and the sale probability of existing properties within the same apartment complex. As I increase the threshold M, the treatment group becomes smaller while the control group expands, resulting in larger estimation samples. Across specifications, the estimated effect on weekly buyer visits ranges from -0.010 to -0.039, broadly comparable to the building-level estimates of -0.037 to -0.044. The estimated effect on the weekly probability of sale ranges from -0.36 to -1.05 percentage points. The estimates suggest that congestion extends beyond individual buildings and can be substantially stronger at the apartment-complex level, particularly when multiple new listings enter simultaneously.

Overall, I find that the apartment-complex-level estimates are qualitatively consistent with the baseline building-level results. These findings provide additional evidence that the congestion externality generated by new listings is not confined to properties within the same building but extends to competing properties in other buildings within the same apartment complex. The results also show that multiple new-listing arrivals within the same week generate stronger congestion effects on incumbent properties.



(a) Number of Visits

(b) Number of Buyers

Figure 6: The Dynamic Effects of New Listings on Buyer Entry at Building Level

Notes: The figure shows the dynamic treatment effects of new listings shock within the same building on number of visits and number of buyers. *Number of Total Visits* is defined as the total number of property visits received by all listings within a building in a given week. *Number of Total Buyers* is the number of unique buyers associated with these visits in the same building-week. The shaded areas represent 95% confidence intervals.

4.4 Buyer Entry and Stock–Flow Matching

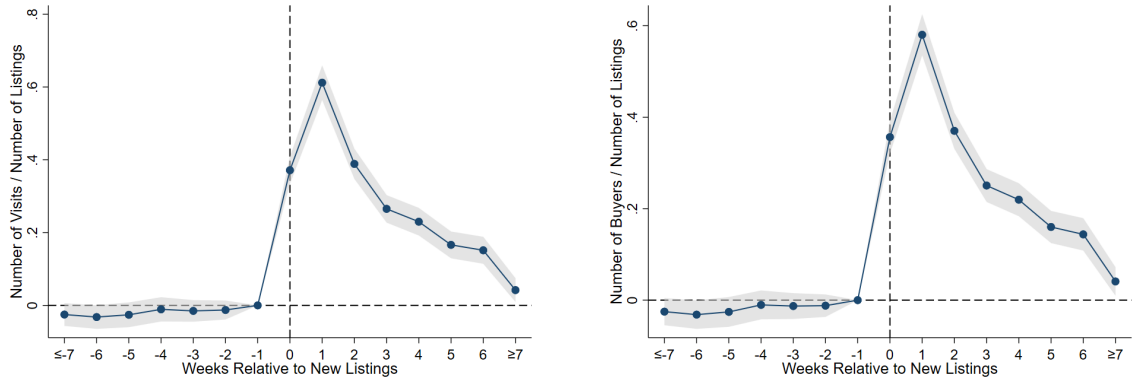
The preceding analysis focuses on incumbent sellers, abstracting from potential responses on the buyer side. Following the discussion in Section 2.4.2, I next examine whether new-listing entry attracts additional buyers into the local market. This distinction is important because the congestion effect may either reflect a fixed amount of buyer search being mechanically divided across more properties or arise even as new listings attract additional buyers. To distinguish between these possibilities, I measure buyer activity at the building-week level using the total number of property visits and the number of unique buyers generating these visits. Figure 6 provides visual evidence that new-listing entry increases both measures of buyer activity. Table A9 shows that new-listing entry increases both weekly visits and unique buyers by approximately 0.6 per building-week. Thus, new listings do more than redistribute a fixed pool of buyer search: they also attract additional buyers into the building.

The arrival of new listings mechanically increases the stock of available properties, as shown in Figure A6. I further examine whether the increase in buyer activity keeps pace with this expansion in housing inventory. Figure 7 shows tha market tightness, measured as visits or unique buyers per active listing, also rises following a new-listing arrival. Thus, new-listing entry increases buyer activity relative to the number of properties available for sale. Yet, despite this increase in market tightness, incumbent properties receive fewer visits and become less likely to sell. This contrast suggests that the additional buyer activity is disproportionately directed toward newly listed properties rather than distributed evenly across sellers.

To understand why new listings attract additional buyer activity without benefiting incumbent sellers, I examine changes in the characteristics of buyers visiting the building. In particular, I investigate whether the additional activity comes from buyers newly entering the housing market or those already searching. Figure 8 presents an event study of buyers' average time on market at the time of their visits to the building, showing a significant but temporary increase following new-listing entry. This suggests that buyers with longer search histories account for a larger share of visits to the building following new-listing entry, indicating that some of the additional activity comes from buyers who have already been searching for some time.

This pattern is consistent with stock–flow matching, whereby new listings initially encounter the existing stock of searching buyers, while older listings increasingly rely on the flow of newly arriving buyers (Gilbukh, 2023). Supporting this interpretation, Figure A5 shows that newly listed properties tend to attract buy-

ers with longer time on market, while older listings have a lower average time on market among their visiting buyers. Figure A10 also shows that buyer visits and sale hazards decline rapidly with listing age. Together, these findings suggest that new listings can attract additional buyer activity into a building by drawing on the existing stock of searching buyers, while redirecting matching opportunities away from incumbent sellers. Consequently, increased buyer activity does not necessarily alleviate incumbent congestion.



(a) Number of Visits / Number of Listings (b) Number of Buyers / Number of Listings

Figure 7: New Listings and Visits and Market Tightness at Building Level

Notes: The figure shows the dynamic treatment effects of a new-listing arrival on market tightness within the same building. Panel (a) measures market tightness as the ratio of the total number of property visits in a building-week to the number of active listings in that building-week. Panel (b) measures market tightness as the ratio of the number of unique buyers in a building-week to the number of active listings in that building-week. The shaded areas represent 95% confidence intervals.

5 Conclusion

This paper studies how new housing supply affects properties already on the market through the buyer search process. I use a sequential-search framework because a standard random-matching model can generate congestion when more sellers compete for a given pool of buyers, but is largely silent about which properties are most exposed to that congestion. Sequential search provides a richer mechanism: new listings enter buyers' search sequences, alter the relative positions of incumbent properties, and thereby change their probabilities of being visited and sold. The model therefore yields additional predictions about the incidence of congestion: stronger effects from more attractive entrants. In addition, stock–flow matching provides a useful framework for understanding buyer–seller matching and predicts stronger congestion effects for recently listed incumbents that still compete with new listings for the existing stock of buyers.

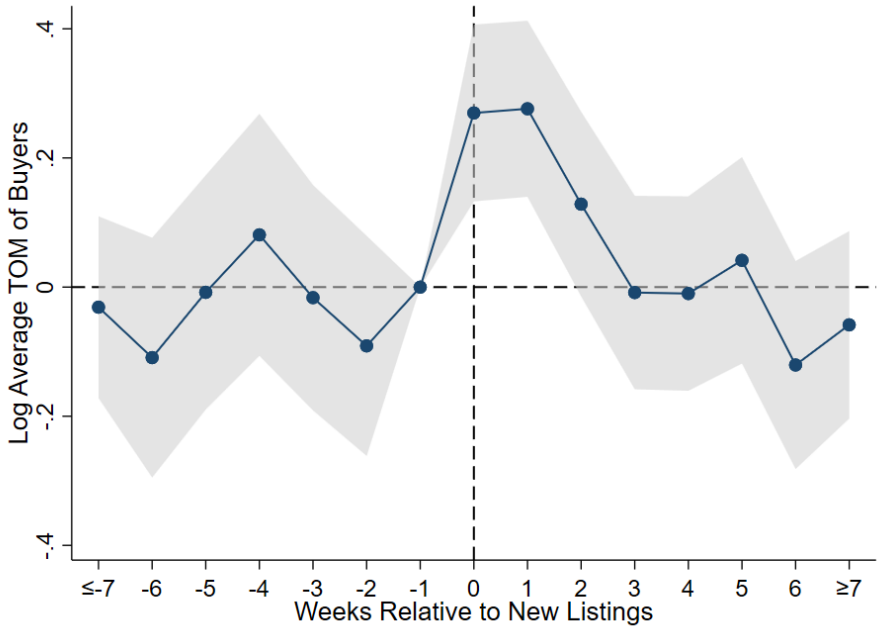


Figure 8: New-Listing Entry and Buyer Time on Market

Notes: The figure reports building-level event-study estimates of the effect of new-listing entry on the log average time on market of buyers visiting properties within the same building. Buyer time on market is measured as the number of days since a buyer began searching at the time of the visit. The shaded areas represent 95% confidence intervals.

The empirical evidence supports these predictions and helps distinguish the search mechanism from a generic increase in seller competition. More price-competitive entrants generate substantially larger congestion effects, and the negative effects are concentrated among relatively recent incumbent listings. The buyer-side evidence further shows that new listings can attract additional buyers into the local market while still reducing visits and sale probabilities for incumbent properties. Thus, congestion is not simply the mechanical consequence of dividing a fixed amount of demand among more sellers; it reflects how new supply changes the ordering and allocation of search across properties.

This perspective also provides a micro-foundation for the stock–flow matching patterns observed in housing markets. Newly listed properties initially interact with the stock of buyers already searching and therefore receive a concentration of visits and matching opportunities shortly after entry. As those opportunities are exhausted, listings increasingly depend on the flow of newly arriving buyers. More broadly, the findings show that housing-market liquidity depends not only on the number of buyers and sellers, but also on where properties appear in buyers' search processes. Understanding this search allocation is therefore important for explaining both the aggregate congestion created by new supply and the unequal incidence of that congestion across incumbent sellers.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author used ChatGPT for language editing and proofreading. The author reviewed and edited the output as needed and takes full responsibility for the content of the published article.

Declaration of Competing Interest

The author declares no conflicts of interest.

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Online Appendix

A Additional Theoretical Results

A.1 Proof of Lemma 1

Proof. The first-ranked house is visited with probability one. For any $m \geq 1$, rank $m+1$ is visited only if the buyer reaches rank m and continues searching afterward.

By Equation (3), $c > 0$ implies

$$z^* < \bar{z},$$

and hence

$$\Pr(z_{ik_m} \geq z^*) > 0.$$

Whenever $z_{ik_m} \geq z^*$,

$$u_{ik_m} \geq r_{ik_m} \geq r_{ik_{m+1}},$$

so the buyer stops after visiting rank m and does not reach rank $m+1$. Thus, conditional on visiting rank m, there is a strictly positive probability of stopping before rank $m+1$. Therefore,

$$\nu_i(m+1) < \nu_i(m),$$

so visiting probability is strictly decreasing in search rank.

□

A.2 Proof of Lemma 2

Proof. Fix buyer i and suppress the buyer subscript. Let A_m denote the event that the buyer wants to purchase the property at rank m at the end of the search process, so that $\Pr(A_m) = \delta_i(m)$.

Consider any realization of match shocks in which the buyer wants to purchase the property at rank $m+1$. Construct a corresponding realization by swapping the shocks at ranks m and $m+1$, leaving all other shocks unchanged. Since match shocks are i.i.d., this swapping preserves their joint probability distribution.

Because properties are ordered by reservation values and the reservation premium z^* is common across properties,

$$r_i(m) \geq r_i(m+1) \quad \Longrightarrow \quad v_i(m) - p(m) \geq v_i(m+1) - p(m+1).$$

After the swap, the property at rank m receives the shock originally assigned to rank $m+1$, and hence

$$v_i(m) - p(m) + z_{m+1} \geq v_i(m+1) - p(m+1) + z_{m+1}.$$

Thus, whenever the buyer wants to purchase the property at rank $m+1$ in the original realization, the corresponding swapped realization makes the buyer want to purchase the property at rank m. Moreover, purchasing a property requires visiting it first. By Lemma 1, rank m is strictly more likely to be visited than rank $m+1$, since it is reached earlier in the search sequence.

Together, the strictly higher probability of being visited, established in Lemma 1, and the weakly higher probability of being chosen for purchase conditional on being visited, established by the swapping argument, imply that

$$\delta_i(m) > \delta_i(m+1), \qquad m = 1, \ldots, S-1.$$

□

A.3 Proof of Proposition 1

Proof. By Lemma 1,

$$\nu_i(m_{ik}) > \nu_i(m_{ik}+1).$$

Hence, equation (12) implies

$$\bar{\nu}_{ik}^1 < \nu_i(m_{ik}).$$

Since the entrant has a positive probability of ranking ahead of property k for at least one buyer, summing across buyers gives

$$V_k^1 < V_k^0.$$

Similarly, Lemma 2 implies

$$\delta_i(m_{ik}) > \delta_i(m_{ik}+1),$$

so equation (13) gives

$$\bar{\delta}_{ik}^1 < \delta_i(m_{ik}).$$

Thus, the probability that property k becomes the most preferred available property of at least one buyer declines:

$$D_k^1 < D_k^0.$$

Suppose property k is sold whenever at least one buyer wants to purchase it, with rationing determining the buyer if multiple buyers are interested. Under this market-clearing assumption,

$$\Pi_k = D_k.$$

It follows immediately that

$$\Pi_k^1 < \Pi_k^0.$$

□

B Additional Tables

Table A1: Balance Test

	Treatment		Control		Difference	
Variable	N	Mean	N	Mean	Treat.–Control	Std. Diff.
Living Rooms	10,355	1.273	7,882	1.276	-0.003	-0.006
Bedrooms	10,355	2.373	7,882	2.428	-0.055***	-0.068
Bathrooms	10,355	1.304	7,882	1.338	-0.034***	-0.062
Log Floor Area	10,355	4.478	7,882	4.495	-0.017**	-0.038
Floor Level	10,355	4.693	7,882	4.401	0.293***	0.086
Log Initial List Price	10,355	14.873	7,882	14.894	-0.021**	-0.037

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. This table compares observable housing characteristics between properties in the treatment and control groups. The difference is calculated as the treatment-group mean minus the control-group mean. Statistical significance is based on two-sample Welch t-tests of equality of means. The standardized difference is calculated as the difference in group means divided by the square root of the average of the two group variances.

Table A2: Summary Statistics for Full Building-week Panel

Variable	N	Mean	Std. Dev.	Min	Max
Visits	2,628,275	0.145	0.841	0	36
Sale Indicator	2,628,275	0.546	7.370	0	100
Log List Price	2,628,275	14.937	0.538	13.122	16.760
Cumulative New Listings	2,628,275	4.278	7.024	0	150
Log Sale Price	6,637	14.840	0.453	13.874	16.380

Notes: The table reports summary statistics for the full building-level panel. Log sale price is observed only for properties that are sold during the sample period. *Cumulative New Listings* measures the total number of new listings that have entered the building up to and including the current week.

Table A3: Summary Statistics under Alternative Treatment Thresholds

	Treatment Threshold (M)					
Variable	1	2	3	4	5	6
Visits	0.160	0.275	0.306	0.330	0.345	0.355
	(0.793)	(1.081)	(1.158)	(1.226)	(1.268)	(1.299)
Sale Indicator	7.481	10.544	10.634	10.743	10.774	10.858
	(26.309)	(30.713)	(30.827)	(30.966)	(31.005)	(31.112)
New Listing	0.440	0.260	0.198	0.151	0.129	0.109
	(0.496)	(0.439)	(0.398)	(0.358)	(0.335)	(0.311)
Log Sale Price	14.820	14.995	14.993	14.991	14.984	14.975
	(0.408)	(0.498)	(0.501)	(0.499)	(0.493)	(0.490)
Observations	31,612	357,561	927,948	1,568,800	2,164,754	2,633,024
Sale Price Observations	95	3,276	10,410	18,800	26,815	33,542

Notes: The table reports means and standard deviations (in parentheses) for alternative treatment thresholds $M = 1, \ldots, 6$. A treatment event occurs when at least M new listings enter a apartment complex within a week. Each sample retains complexes with exactly one treatment event or no treatment events. Visits and the sale indicator are measured at the property-week level. The sale indicator equals 100 in the week of sale and zero otherwise. Log sale price is observed only for sold properties.

Table A4: New-Listing Entry and Sale Probability Conditional on Buyer Visits

	TWFE			Sun and Abraham		
	(1)	(2)	(3)	(4)	(5)	(6)
New Listing	-0.060	-0.047	-0.076*	-0.108	-0.103	-0.174**
	(0.037)	(0.038)	(0.045)	(0.076)	(0.077)	(0.088)
Buyer Visits	1.636***	1.628***	1.649***	1.637***	1.628***	1.650***
	(0.071)	(0.070)	(0.074)	(0.071)	(0.070)	(0.074)
N	487,200	487,170	459,273	487,200	487,170	459,273
R^2	0.110	0.124	0.256	0.111	0.124	0.256
Housing Attributes	✓	✓	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighborhood × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. The dependent variable is the weekly sale indicator, equal to 100 in the week a property is sold and zero otherwise. Columns (1)–(3) report TWFE estimates controlling for weekly buyer visits. Columns (4)–(6) report interaction-weighted estimates following Sun and Abraham (2021), with buyer visits included as a covariate. Housing attributes and the indicated fixed effects are included in all specifications. Standard errors are clustered at the building level and reported in parentheses.

Table A5: The Impact of Cumulative New Listings on Visits and Sales

	Number of Visits			Sale Probability		
	(1)	(2)	(3)	(4)	(5)	(6)
Cumulative New Listings	-0.003***	-0.001***	-0.004***	-0.022***	-0.018***	-0.026***
	(0.000)	(0.000)	(0.001)	(0.003)	(0.002)	(0.003)
N	2,628,275	2,628,245	2,609,218	2,628,275	2,628,245	2,609,218
R^2	0.130	0.136	0.177	0.048	0.051	0.101
Housing Attributes	✓	✓	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E	✓	✓	✓	✓	✓	✓
Neighbor $\times$ Week F.E.		✓			✓	
Complex $\times$ Week F.E.			✓			✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. *Cumulative New Listings* is defined as the total number of new listings that have entered the building from the beginning of the sample period up to and including the current week. The dependent variables are the number of visits received by the property in a given week and an indicator for whether the property is sold in that week. Control variables include floor level, floor area, and the numbers of kitchens, bedrooms, and living rooms.

Table A6: The Impact of Cumulative New Listings on Prices

	List Prices			Sale Prices		
	(1)	(2)	(3)	(4)	(5)	(6)
Cumulative New Listings	-0.000	0.000	0.000	0.000	0.001	-0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.003)
N	2,628,275	2,628,245	2,609,218	6,637	5,030	1,831
R^2	0.979	0.979	0.979	0.979	0.985	0.983
Housing Attributes	✓	✓	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓	✓	✓
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E	✓	✓	✓	✓	✓	✓
Neighbor $\times$ Week F.E.		✓			✓	
Complex $\times$ Week F.E.			✓			✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. *Cumulative New Listings* is defined as the total number of new listings that have entered the building from the beginning of the sample period up to and including the current week. The dependent variables are list prices and sale prices of the property. Control variables include floor level, floor area, and the numbers of kitchens, bedrooms, and living rooms.

Table A7: The Impact of New Listings on Visits at the Apartment Complex Level

$M =$	Treatment: At Least M New Listings at Complex-Week					
	1	2	3	4	5	6
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-0.003	-0.017	-0.036***	-0.024***	-0.036***	-0.037***
	(0.031)	(0.013)	(0.010)	(0.009)	(0.008)	(0.009)
N	31,612	357,561	927,948	1,568,800	2,164,754	2,633,024
R^2	0.329	0.168	0.148	0.141	0.137	0.135
Housing Attributes	✓	✓	✓	✓	✓	✓
Complex F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighbor $\times$ Week F.E.	✓	✓	✓	✓	✓	✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. This table reports the effects of new listings on buyer visits at the apartment complex level. In column M, a complex-week is classified as treated if at least M new properties are listed within the complex during that week, while complex-weeks with fewer than M new listings serve as the control group.

Table A8: The Impact of New Listings on Sales at the Apartment Complex Level

$M =$	Treatment: At Least M New Listings at Complex-Week					
	1	2	3	4	5	6
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment	-0.720	-0.523	-0.275	-0.989***	-0.957***	-1.006***
	(1.171)	(0.465)	(0.312)	(0.288)	(0.281)	(0.275)
N	31,612	357,561	927,948	1,568,800	2,164,754	2,633,024
R^2	0.549	0.274	0.222	0.205	0.196	0.194
Housing Attributes	✓	✓	✓	✓	✓	✓
Time-on-Market F.E.	✓	✓	✓	✓	✓	✓
Complex F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighbor $\times$ Week F.E.	✓	✓	✓	✓	✓	✓

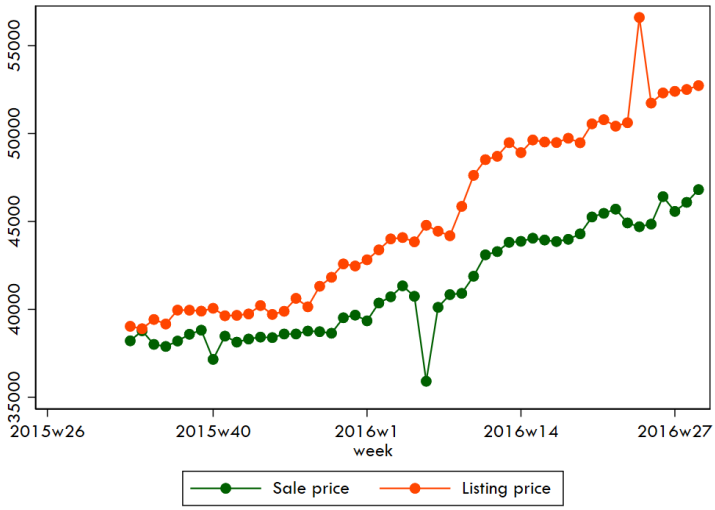
Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. This table reports the effects of new listings at the apartment complex level. In column M, a complex-week is classified as treated if at least M new properties are listed within the complex during that week, while complex-weeks with fewer than M new listings serve as the control group.

Table A9: The Impact of New Listings on Buyer Entry

	Number of Total Visits			Number of Total Buyers		
	(1)	(2)	(3)	(4)	(5)	(6)
New Listing	0.625***	0.640***	0.682***	0.601***	0.615***	0.653***
	(0.019)	(0.019)	(0.023)	(0.018)	(0.018)	(0.021)
N	321,725	321,695	273,851	321,725	321,695	273,851
R^2	0.232	0.255	0.430	0.235	0.257	0.431
Building F.E.	✓	✓	✓	✓	✓	✓
Week F.E.	✓	✓	✓	✓	✓	✓
Neighbor × Week F.E.		✓			✓	
Complex × Week F.E.			✓			✓

Notes: *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the building level. *Number of Total Visits* is defined as the total number of property visits received by all listings within a building in a given week. *Number of Total Buyers* is the number of unique buyers associated with these visits in the same building-week.

C Additional Figures



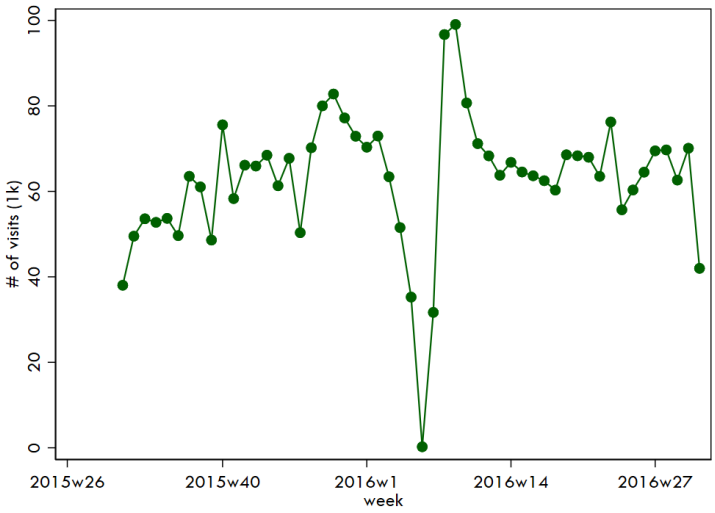
(a) Price Dynamics



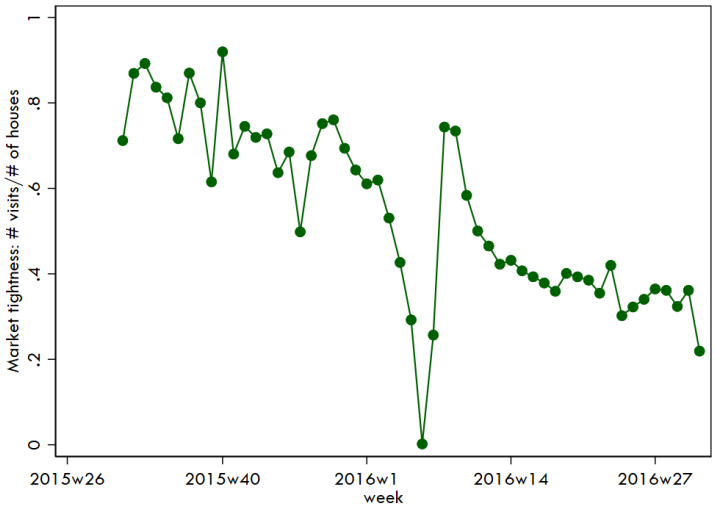
(b) New Listings and Sales

Figure A1: The Prices, Listings, and Sales Dynamics

Notes: Panel (a) shows the evolution of average transaction prices and list prices in Beijing over the sample period. Panel (b) shows the corresponding dynamics in the number of new listings and housing transactions.



(a) Number of Visits



(b) Market Tightness

Figure A2: The Visits and Market Tightness Dynamics

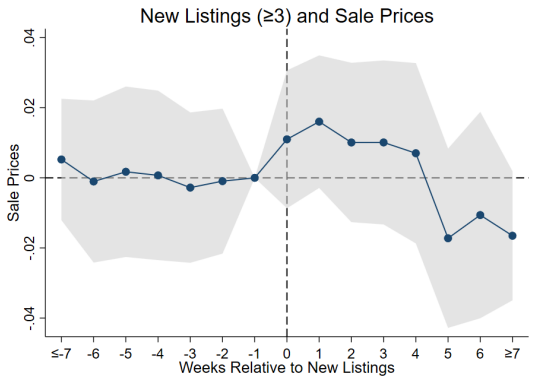
Notes: Panel (a) shows the evolution of the total number of buyer visits in each week over the sample period. Panel (b) shows the corresponding evolution of market tightness, measured as the ratio of total buyer visits to the number of active listings.



(a) At Least 1 New Listing



(b) At Least 2 New Listings



(c) At Least 3 New Listings



(d) At Least 4 New Listings



(e) At Least 5 New Listings



(f) At Least 6 New Listings

Figure A3: The Effects of at Least M New Listings on Sale Prices

Notes: The figure shows how sale prices of sold properties changes after new listings shock within the same apartment complex. The six panels display the results for different treatment group definitions, where the number of new listings is at least 1, at least 2, at least 3, at least 4, at least 5, and at least 6, respectively. The shaded areas represent 95% confidence intervals.

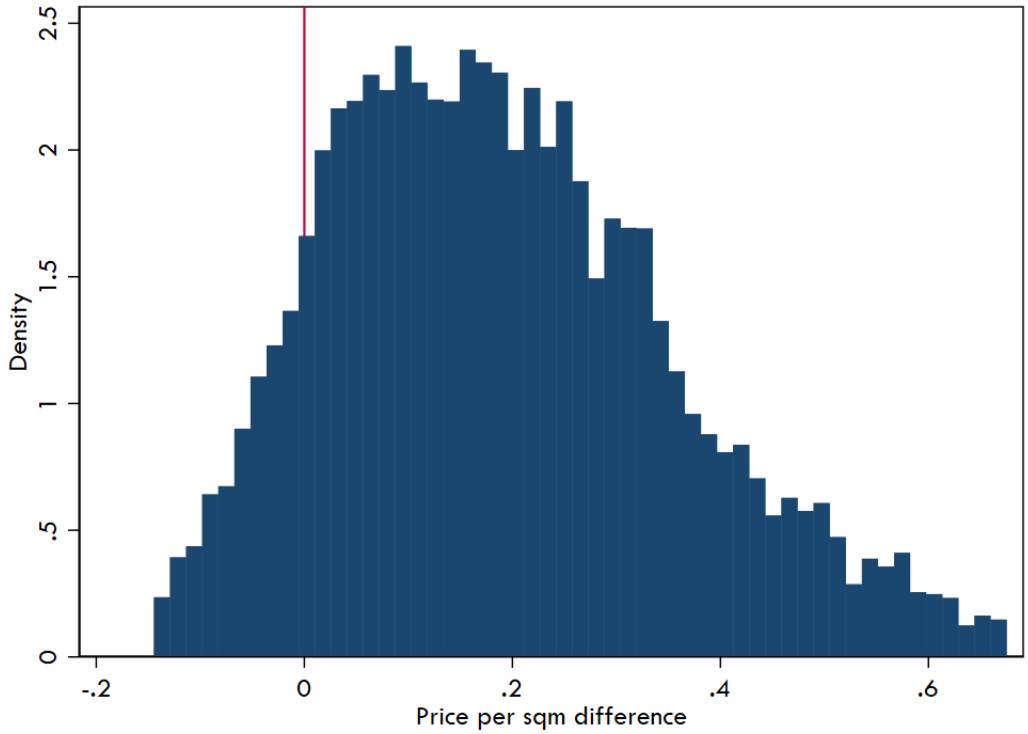
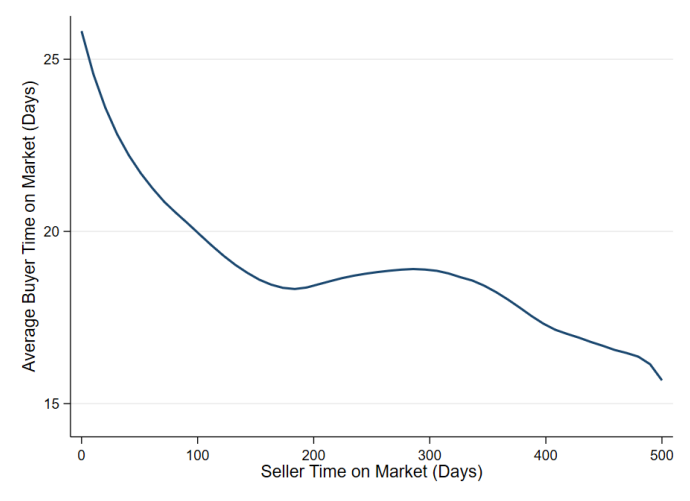
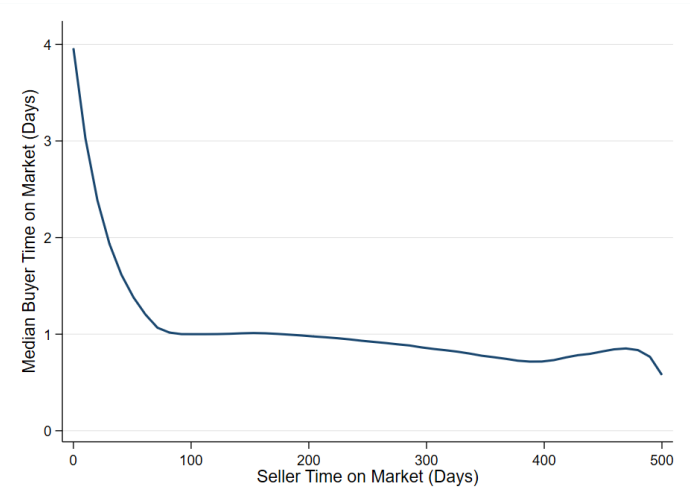


Figure A4: Histogram of the Percentage Price Difference

Notes: This figure is a histogram of the percentage price difference between new listing houses and existing houses. This chart shows that a large number of new listings have a higher price per square meter than existing listings, up to 70%. A small number of new listings have a lower price per square meter than existing listings, up to 20%. New listings entering the market are generally more expensive primarily because prices have been rising across time for the whole market (see Figure A1).



(a) Mean Buyer Time on Market



(b) Median Buyer Time on Market

Figure A5: Stock–Flow Matching in Housing Search

Notes: The figure illustrates the relationship between seller time on market and the search duration of buyers visiting the property. Panel (a) plots the mean time on market of visiting buyers against seller time on market, while Panel (b) reports the corresponding median. Buyer time on market is measured as the number of days since the buyer entered the market at the time of the visit. Both panels use local polynomial smoothing.

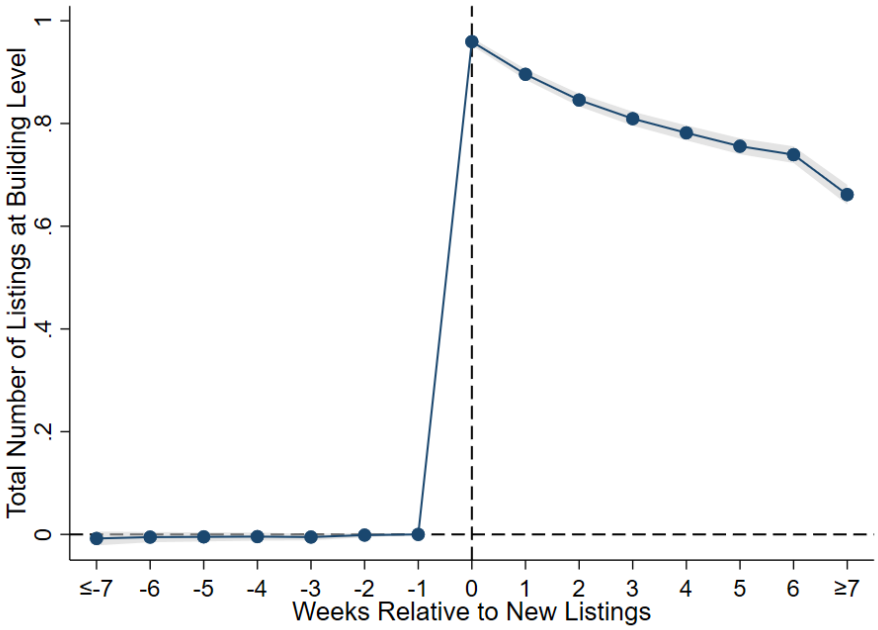
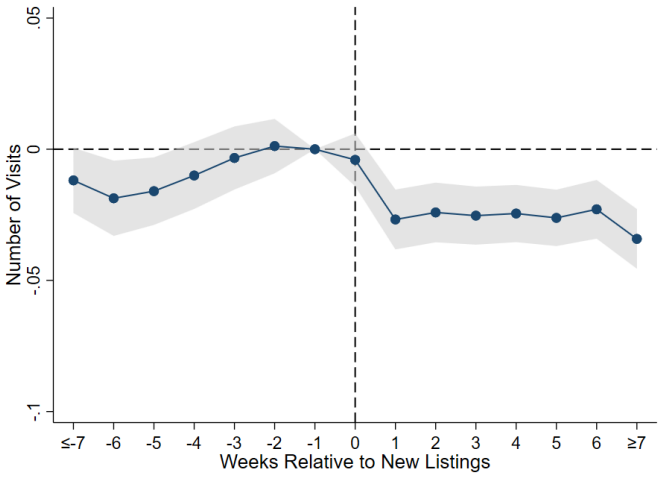
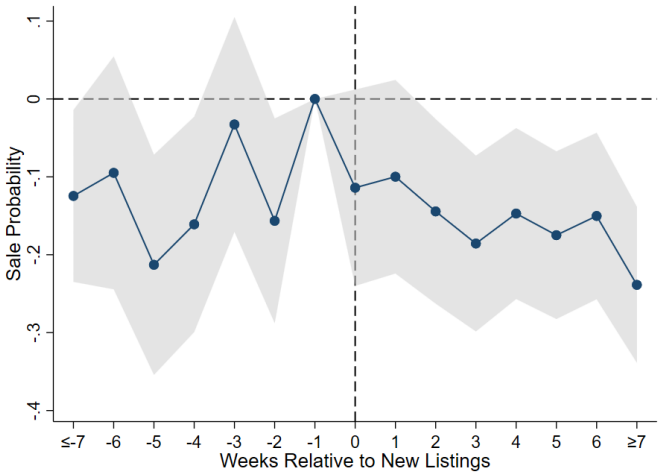


Figure A6: New Listings and Total Listings at Building Level

Notes: The figure shows the dynamic treatment effects of a new-listing arrival on the total number of active housing listings within the same building. Following the arrival, the stock of active for-sale listings mechanically increases by one and gradually declines as the new listing may subsequently be sold. The shaded areas represent 95% confidence intervals.



(a) Number of Visits / Number of Listings



(b) Number of Buyers / Number of Listings

Figure A7: Dynamic Effects of the First New-Listing Arrival

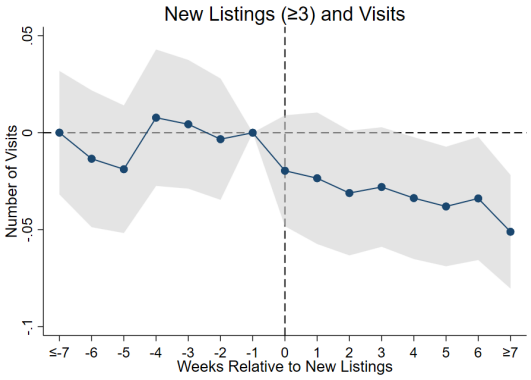
Notes: The figure reports event-study estimates using the full building-level panel. Treatment timing is defined as the week in which a building experiences its first new-listing arrival during the sample period. Panel (a) reports the effects on weekly buyer visits to incumbent properties, while Panel (b) reports the effects on their weekly probability of sale. Because many buildings experience their first new-listing arrival near the beginning of the observation window, the number of available pre-treatment observations is limited for some event times. The shaded areas represent 95% confidence intervals.



(a) At Least 1 New Listing



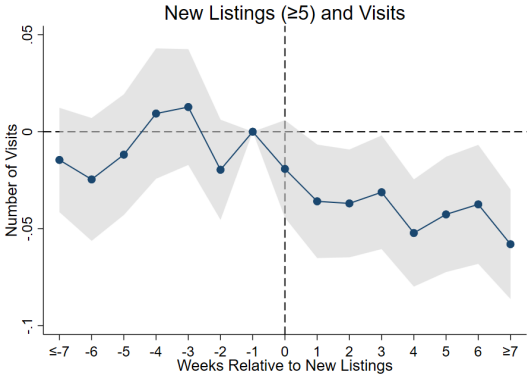
(b) At Least 2 New Listings



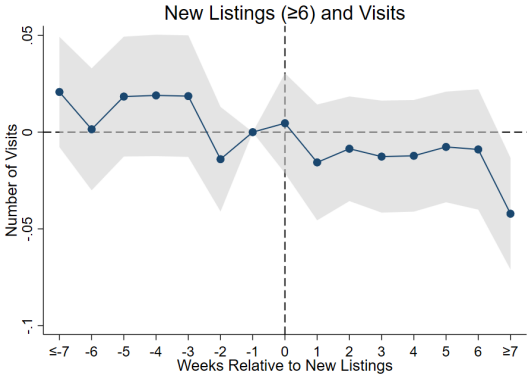
(c) At Least 3 New Listings



(d) At Least 4 New Listings



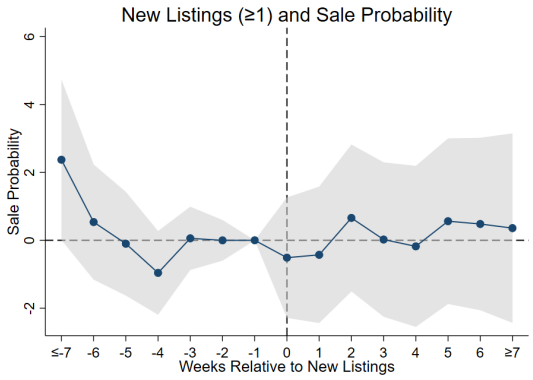
(e) At Least 5 New Listings



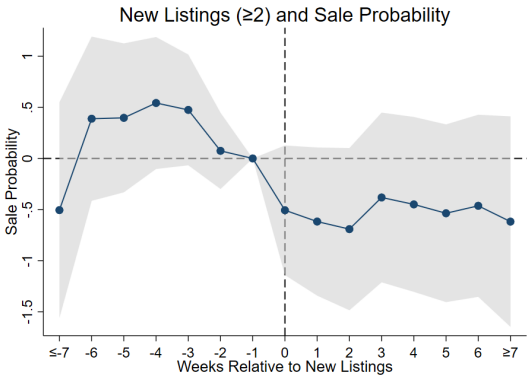
(f) At Least 6 New Listings

Figure A8: The Effects of at Least M New Listings on Visits

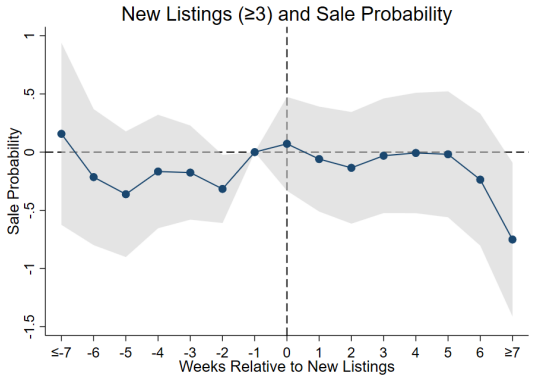
Notes: The figure shows how number of visits of the existing listings changes after new listings shock within the same apartment complex. The six panels display the results for different treatment group definitions, where the number of new listings within one week is at least 1, at least 2, at least 3, at least 4, at least 5, and at least 6, respectively. The shaded areas represent 95% confidence intervals.



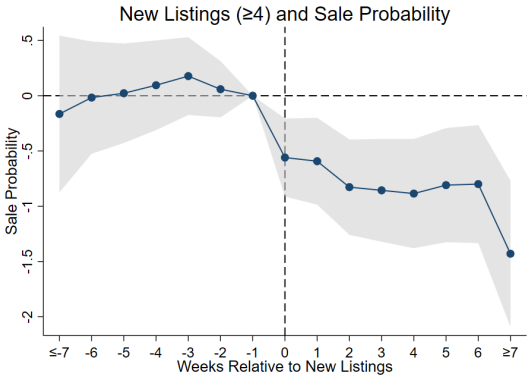
(a) At Least 1 New Listing



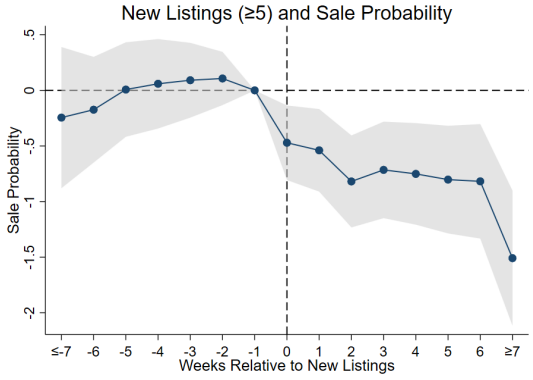
(b) At Least 2 New Listings



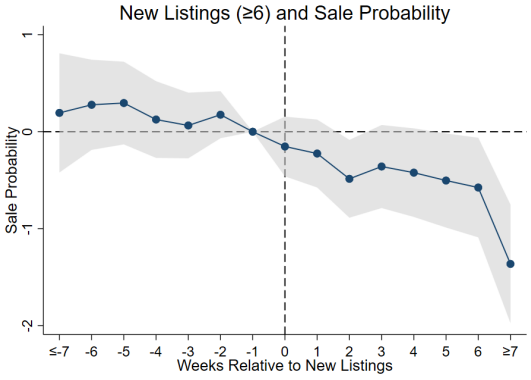
(c) At Least 3 New Listings



(d) At Least 4 New Listings



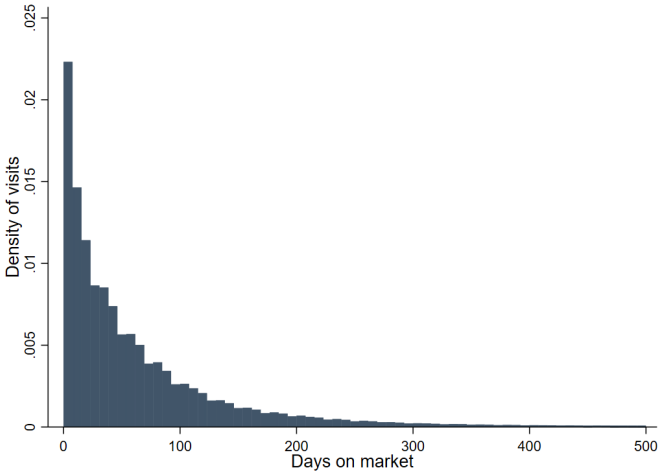
(e) At Least 5 New Listings



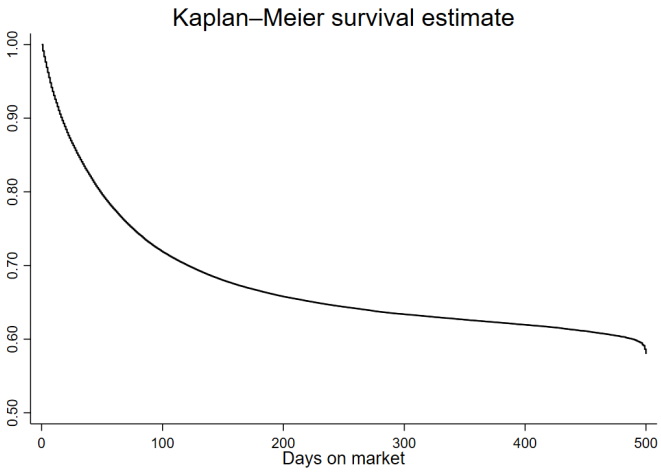
(f) At Least 6 New Listings

Figure A9: The Effects of at Least M New Listings on Sale Probability

Notes: The figure shows how transaction probability of the existing listings changes after new listings shock within the same apartment complex. The six panels display the results for different treatment group definitions, where the number of new listings within one week is at least 1, at least 2, at least 3, at least 4, at least 5, and at least 6, respectively. The shaded areas represent 95% confidence intervals.



(a) Density of Visits Over Time-on-Market



(b) Survival Rate Over Time-on-Market

Figure A10: Visits and Survival Rate Over Time-on-Market

Notes: Panel (a) shows the distribution of all visits over sellers' time-on-market. Panel (b) presents Kaplan–Meier survival estimates for property listings, where a sale is defined as the failure event. All properties enter the risk set at the time of listing and are right-censored at the end of August 2016, allowing us to characterize the time-to-sale process and the corresponding sale hazard.